



Drug prescribing for older adults

AUTHOR: [Paula A Rochon, MD, MPH, FRCPC](#)

SECTION EDITOR: [Kenneth E Schmader, MD](#)

DEPUTY EDITOR: [Jane Givens, MD, MSCE](#)

All topics are updated as new evidence becomes available and our [peer review process](#) is complete.

Literature review current through: **Apr 2026**.

This topic last updated: **Mar 03, 2026**.

INTRODUCTION

Optimizing drug therapy is an essential part of caring for an older person. The process of prescribing a medication is complex and includes: deciding that a drug is indicated, choosing the best drug, determining a dose and schedule appropriate for the patient's physiologic status, monitoring for effectiveness and toxicity, educating the patient about expected side effects, and indications for seeking consultation.

Avoidable adverse drug events (ADEs) are the serious consequences of inappropriate drug prescribing. The possibility of an ADE should always be borne in mind when evaluating an older adult; any new symptom should be considered drug-related until proven otherwise.

Prescribing for older patients presents unique challenges. Premarketing drug trials often exclude older adults, and approved doses may not be appropriate for older adults [1]. Many medications need to be used with special caution because of age-related changes in pharmacokinetics (ie, absorption, distribution, metabolism, and excretion) and pharmacodynamics (the physiologic effects of the drug).

Particular care must be taken in determining drug doses when prescribing for older adults. An increased volume of distribution may result from the proportional increase in body fat relative to skeletal muscle with aging. Decreased drug clearance may result from the natural decline in kidney function with age, even in the absence of renal disease [2]. Larger drug storage reservoirs and decreased clearance prolong drug half-lives and lead to increased plasma drug concentrations in older people.

As examples, the volume of distribution for [diazepam](#) is increased, and the clearance rate for [lithium](#) is reduced, in older adults. The same dose of either medication would lead to higher plasma concentrations in an older, compared with a younger patient. Also, from the

pharmacodynamic perspective, increasing age may result in an increased sensitivity to the effects of certain drugs, including benzodiazepines [3-6] and opioids [7].

Hepatic function also declines with advancing age, and age-related changes in hepatic function may account for significant variability in drug metabolism among older adults [8]. Especially when polypharmacy is a factor, decreasing hepatic function may lead to adverse drug reactions (ADRs).

A stepwise approach to optimized prescribing of drug therapy for older adults will be reviewed here. Drug treatments for specific conditions in the older population are discussed separately.

MEDICATION USE BY OLDER ADULTS

Medications (prescription, over-the-counter, and herbal preparations) are widely used by older adults.

Prescription medications — A survey in the United States of a representative sampling of 2206 community-dwelling adults (aged 62 through 85 years) was conducted by in-home interviews and use of medication logs between 2010 and 2011 [9]. At least one prescription medication was used by 87 percent. Five or more prescription medications were used by 36 percent, and 38 percent used over-the-counter medications.

In a sample of Medicare beneficiaries discharged from an acute hospitalization to a skilled nursing facility, patients were prescribed an average of 14 medications, including over one-third with adverse effects that could exacerbate underlying geriatric syndromes [10].

Herbal and dietary supplements — Use of herbal or dietary supplements (eg, ginseng, ginkgo biloba extract, and glucosamine) by older adults has been increasing, from 14 percent in 1998 [11] to 63 percent in 2010 [9]. In one study of over 3000 ambulatory adults ages 75 years or older, almost three-quarters used at least one prescription drug and one dietary supplement [12]. Often, clinicians do not question patients about use of herbal medicines, and patients do not routinely volunteer this information. In one United States survey, three-quarters of respondents aged 18 years and older reported that they did not inform their clinician that they were using unconventional medications [13].

Herbal medicines may interact with prescribed drug therapies and lead to adverse events, underscoring the importance of routinely questioning patients about the use of unconventional therapies. Examples of herbal-drug therapy interactions include ginkgo biloba extract taken with [warfarin](#), causing an increased risk of bleeding, and St. John's wort taken with serotonin-reuptake inhibitors, increasing the risk of serotonin syndrome in older adults [14]. A study of the use of 22 supplements in a survey of 369 patients aged 60 to 99 years

found potential interactions between supplements and medications for 10 of the 22 supplements surveyed [15]. (See "[Overview of herbal medicine and dietary supplements](#)", [section on 'Herb-drug interactions'](#).)

Many older adults receive their information about herbal products from the internet. Eighty percent of 338 retail web sites identified in a search of the eight most widely used herbal supplements (ginkgo biloba, St. John's wort, echinacea, ginseng, garlic, saw palmetto, kava, and valerian root) made at least one health claim suggesting that the therapy could treat, prevent, or even cure specific conditions [16].

QUALITY MEASURES OF DRUG PRESCRIBING

Multiple factors contribute to the appropriateness and overall quality of drug prescribing. These include avoidance of inappropriate medications, appropriate use of indicated medications, monitoring for adverse effects and drug levels, avoidance of drug-drug interactions, and involvement of the patient and integration of patient values [17].

Measures of the quality of prescribing often focus on one or some of these factors, but rarely on all. Furthermore, the predictive value of these measures of "quality of prescribing" in determining important long-term outcomes of care have not been determined. Approaches to decrease inappropriate prescribing in older adults include educational interventions, peer comparison feedback, computerized order entry and decision support, multidisciplinary team care led by physicians, clinical pharmacists, and combinations of these approaches [18]. Available data for these interventions generally show significant improvements in inappropriate prescribing but mixed results for health outcomes or costs [17,19]. A 2016 systematic review of eight studies of different prescribing interventions in the long-term care setting (medication review, case conferences, staff education, clinical decision support technology, and/or some combination of these) showed no effect of the interventions on hospital admissions, adverse drug events (ADEs), or mortality [19]. A previous 2008 systematic review of 10 studies of computerized physician order entry with clinical decision support in the hospital and ambulatory setting showed a mixed effect on reduction in ADEs, with five studies that showed a statistically significant reduction in ADEs, four that showed nonsignificant decrease, and one study that showed no impact on rate of ADEs [20].

POLYPHARMACY

Polypharmacy is defined simply as the use of multiple medications by a patient. The precise minimum number of medications used to define "polypharmacy" is variable, but generally ranges from 5 to 10 [21]. While polypharmacy most commonly refers to prescribed medications, it is important to also consider the number of over-the-counter and

herbal/supplements used. Problematic polypharmacy is defined as the use of multiple medications in a way that is not considered to be appropriate [22].

The issue of polypharmacy is of particular concern in older people who, compared with younger individuals, tend to have more disease conditions for which therapies are prescribed. It has been estimated that 20 percent of Medicare beneficiaries have five or more chronic conditions and 50 percent receive five or more medications [23]. Among ambulatory older adults with cancer, 84 percent were receiving five or more and 43 percent were receiving 10 or more medications, in one study [24].

The use of greater numbers of drug therapies has been independently associated with an increased risk for an adverse drug event (ADE), irrespective of age [25], and increased risk of hospital admission [26,27]. Polypharmacy has also been associated with decreased physical and cognitive capability, even after adjusting for disease burden [28]. It is difficult to eliminate the impact of confounding factors in considering the relationship between polypharmacy and a variety of outcomes in observational studies [29].

There are multiple reasons why older adults are especially impacted by polypharmacy:

- Older individuals are at greater risk for ADEs due to metabolic changes and decreased drug clearance associated with aging; this risk is compounded by increasing numbers of drugs used.
- Polypharmacy increases the potential for drug-drug interactions and for prescription of potentially inappropriate medications [30].
- Polypharmacy was an independent risk factor for hip fractures in older adults in one case-control study, although the number of drugs may have been an indicator of higher likelihood of exposure to specific types of drugs associated with falls (eg, central nervous system [CNS]-active drugs) [31].
- Polypharmacy increases the possibility of "prescribing cascades" [32]. A prescribing cascade develops when an ADE is misinterpreted as a new medical condition and additional drug therapy is then prescribed to treat this medical condition (see '[Prescribing cascades](#)' below). Prescribing cascades are part of the definition of problematic polypharmacy.
- Use of multiple medications can lead to problems with adherence in older adults, especially if compounded by visual or cognitive impairment. A 2017 systematic review of observational studies suggested that drug regimen complexity is associated with medication nonadherence [26].

A balance is required between over- and under-prescribing. Multiple medications are often required to manage clinically complex older adults. Clinicians are often challenged with the need to match the complex needs of their older patients with those of disease-specific clinical

A more systematic approach is required to guide the tailoring of medication regimens to the needs of individuals. One important principle is to match the medication regimen to the patient's condition and goals of care. This includes a careful consideration of the medications that should be discontinued or substituted [34] ([table 1](#)).

It is particularly important to reconsider medication appropriateness late in life. A model for appropriate prescribing for patients late in life has been proposed [35] ([table 2](#)). The process considers the patients' remaining life expectancy and the goals of care in reviewing the need for existing medications and in making new prescribing decisions. For example, if a patient's life expectancy is short and the goals of care are palliative, then prescribing a prophylactic medication requiring several years to realize a benefit may not be considered appropriate. This is increasingly being recognized as an important consideration when managing individuals with advanced dementia [36]. Additionally, therapeutic medications (eg, antibiotics for pneumonia) may not increase comfort or quality of life when palliative care is the objective [37].

[illegible]

Table 1 - larger image below

POTENTIALLY INAPPROPRIATE MEDICATIONS

Various criteria have been developed by expert panels in Canada, the United States, and Europe [38-43], to assess the quality of prescribing practices and medication use in older adult individuals. The most widely used criteria for inappropriate medications are the Beers criteria. (See 'Beers criteria' below.)

In another approach, a Drug Burden Index has been modelled incorporating drugs with anticholinergic or sedative effects, total number of medications, and daily dosing [44,45]. An increased drug burden for anticholinergic and sedative medications was associated with impaired performance on mobility and cognitive testing in high-functioning community-based older adults. [Zolpidem](#), in particular, was implicated in 21 percent of emergency department visits for adverse drug events (ADEs) related to psychiatric medication among adults 65 years and older [46].

Total number of medications was not associated with impaired performance when sedatives and anticholinergics were excluded [44,45]. A high Drug Burden Index has been correlated with increased risk for functional decline in community dwellers [45] and with increased risk of falls in residents in long-term care facilities [47].

The global prevalence of potentially inappropriate medication (PIM) use in outpatient services was evaluated in a 2023 meta-analysis of 94 studies representing over 371 million older adults in 17 countries [48]. The pooled prevalence of PIM use was 36.7 percent, with benzodiazepines being the most common PIM. PIM use has increased over the past 20 years, especially in high-income countries.

Anticholinergic activity — Anticholinergic medications are associated with multiple adverse effects to which older individuals are particularly susceptible. Nonetheless, an analysis of United States medication expenditures between 2005 and 2009 found that 23.3 percent of community-dwelling persons >65 years with dementia were prescribed medications with clinically significant anticholinergic activity (AA) [49].

Adverse effects associated with anticholinergic use in older adults include memory impairment, confusion, hallucinations, dry mouth, blurred vision, constipation, nausea, urinary retention, impaired sweating, and tachycardia. A case-control study found an association between anticholinergic use and risk of community-acquired pneumonia [50]. Anticholinergics can precipitate acute glaucoma episode in patients with narrow angle glaucoma and acute urinary retention in patients with benign prostatic hypertrophy. Specific studies of the relationship between dementia and anticholinergic use include the following:

- In a population study of 6912 adults 65 years and older, those taking anticholinergic drugs were at increased risk for cognitive decline and dementia and risk decreased with medication discontinuation [51].
- In a population of 3434 adults age 65 and older in one health care setting, who had no baseline dementia and who were followed for 10 years, the risk of dementia and Alzheimer's disease increased in a dose-response relationship with use of anticholinergic drug classes (primarily first-generation antihistamines, tricyclic antidepressants, and bladder antimuscarinics) [52].
- In another population of 13,004 individuals aged 65 and older, use of anticholinergic medications was also shown to be associated with greater decline in cognition as measured by the Mini-Mental State Examination [53]. In addition, anticholinergic medication use was associated with increased mortality over a two-year period after adjustment for multiple factors, including comorbid health conditions.

At usual doses, AA is most significantly elevated for [amitriptyline](#), [atropine](#), [clozapine](#), [dicyclomine](#), [doxepin](#), L-hyoscyamine, [thioridazine](#), and [tolterodine](#) [54]. AA also was

increased for [chlorpromazine](#), [diphenhydramine](#), [nortriptyline](#), [olanzapine](#), [oxybutynin](#), and [paroxetine](#). It is important to recognize that higher doses of an agent with low or moderate AA can produce significant AA effects, and, similarly, the cumulative effects of multiple agents with low AA can produce significant AA effects. A listing of medication classes that contain significant AA is shown in a table ([table 3](#)).

Alternative drugs with lower AA are available in many classes represented by these drugs. However, adverse drug reactions (ADRs) other than AA should also be taken into account in weighing the clinical benefits of possible substitutions (eg, dyskinesias and sedation with [haloperidol](#) and [perphenazine](#)).

Multiple scales in addition to the Drug Burden Index have been developed to qualify the AA of medications. In one study, a higher score on each of nine different anticholinergic burden scales was associated with increased risk for hospitalization and length of stay, falls, and medical utilization [55].

Beers criteria — The Beers criteria are a widely cited list of medications considered potentially inappropriate for use in older patients, mostly due to high risk for adverse events [56]. Medications are grouped into five categories: those potentially inappropriate in most older adults, those that should typically be avoided in older adults with certain conditions, drugs to use with caution due to the potential for harmful adverse effects, drug-drug interactions, and drug dose adjustment based on kidney function. A notable limitation of the criteria are that they are most applicable to clinical care in the United States, as they focus on medications available in that market.

The criteria have been repeatedly updated, most recently in 2023 [57], and are available on the American Geriatrics Society [website](#). Selected changes in the 2023 update are described below:

- Avoid use of [rivaroxaban](#) for long-term treatment of nonvalvular atrial fibrillation or venous thromboembolism (VTE), due to higher risk of bleeding in older adults than other direct-acting oral anticoagulants (DOACs).
- Avoid [warfarin](#) as initial therapy for VTE or nonvalvular atrial fibrillation unless alternatives (eg, DOACs) are contraindicated or there are substantial barriers to their use (eg, formulary restrictions).
- Avoid sulfonylureas as first- or second-line monotherapy or add on-therapy due to higher risk of cardiovascular events, all-cause mortality, and hypoglycemia than alternative choices. If a sulfonylurea must be used, then a short-acting agent is preferred.
- Avoid the initiation of oral or transdermal estrogen in older women. Topical vaginal estrogen remains appropriate as treatment for symptomatic vaginal atrophy or urinary

tract infection prophylaxis. Deprescribing should be considered for older women already using nonvaginal estrogen replacement.

- Avoid initiating [aspirin](#) for primary prevention of cardiovascular disease. Consider deprescribing aspirin in older patients already taking it for primary prevention.

While the Beers criteria is supported by evidence, the American Geriatrics Society (AGS) advises that clinicians must consider many factors in prescribing decisions, including using common sense and clinical judgment, understanding that strict adherence to the criteria is not always possible [58]. Further, an evidence-based consensus recommendation from the AGS identifies alternative pharmacologic and nonpharmacologic treatments that can be used to address pruritus, pain, diabetes, insomnia, delirium, gastroesophageal reflux, and recurrent urinary tract infections in place of potentially inappropriate medications [59].

Several studies using older versions of the Beers criteria have identified that use of drugs identified as "inappropriate" was widespread in the United States, Canada, and Europe [60,61]. For example, in a sample of community-dwelling older adults in the United States, 43 percent used at least one medication that would be deemed potentially inappropriate by the criteria, with nonsteroidal antiinflammatory drugs (NSAIDs) being the most common [62].

In a study among 4842 older United States Medicare beneficiaries, there were declines in prescriptions for 2012 and 2019 Beers criteria potentially inappropriate CNS-active prescriptions (antidepressants with strong anticholinergic properties, antipsychotics, barbiturates, benzodiazepines, and nonbenzodiazepine hypnotics) from 2013 to 2021 (19.9 to 16.2 percent) [63]. There were significant declines in prescribing rates for benzodiazepines and nonbenzodiazepine hypnotics, but not for antipsychotics, antidepressants, and barbiturates. Patients with dementia, or cognitive impairment no dementia, were more likely to receive potentially inappropriate medications than those with normal cognition (25.1, 21.7, and 17 percent, respectively).

Some of the inappropriate drug therapies identified on the Beers list are available as over-the-counter products [64]. This reinforces the need to always consider over-the-counter drug therapies when reviewing a patient's medications and to educate individuals on potential problems that can arise from the use of over-the-counter preparations.

The Beer's criteria have been used to monitor quality of care for older adults. Studies of earlier versions of the Beers criteria found that while the criteria did predict some adverse outcomes, results were mixed [65-68].

Other criteria sets — The Screening Tool of Older Person's Prescriptions (STOPP) criteria, another tool for identifying inappropriate prescribing, was introduced in 2008 and most recently updated in 2023 [42,69-71]. In a cluster randomized trial in Ireland, presenting attending physicians with potentially inappropriate medications based on the STOPP/START

(Screening Tool to Alert doctors to the Right Treatment) criteria reduced the number of adverse drug events and medication costs during the index hospitalization, but did not reduce length of stay [72]. In a trial among hospitalized adults in Switzerland comparing the interventions of STOPP/START and potentially inappropriate medication-check (PIM-check), both were found to decrease potentially inappropriate medications, and STOPP/START was also associated with an improvement in activities of daily living [73].

The FORTA (Fit FOR The Aged) list identifies medications rated in four categories (clear benefit; proven but limited efficacy or some safety concerns; questionable efficacy or safety profile, consider alternative; clearly avoid and find alternative) with ratings based on the individual patient's indication for the medication [74]. The tool, developed in Germany, has undergone consensus validation with a panel of geriatricians [75], but studies of its impact on clinical outcomes are ongoing.

A set of criteria specific to the emergency department (the Geriatric Emergency Medication Safety Recommendations) consists of eight classes of potentially inappropriate medications to avoid in older patients [76]. Using these criteria, in a large cross-sectional analysis of patients 65 years or older discharged from an emergency department, potentially inappropriate medications were prescribed at discharge in 6.5 percent of encounters [77]. The most common drug classes were skeletal muscle relaxants and first-generation antihistamines.

Health care financing administration — The Centers for Medicare and Medicaid Services drug utilization review criteria target eight prescription drug classes (digoxin, calcium channel blockers, ACE inhibitors, H2 receptor antagonists, NSAIDs, benzodiazepines, antipsychotics, and antidepressants) and focus on four types of prescribing problems (inappropriate dose, inappropriate duration of therapy, duplication of therapies, and potential for drug-drug interactions). In one study, 19 percent of 2508 community-dwelling older adults were using one or more medications inappropriately; NSAIDs and benzodiazepines were the drug classes with the most potential problems [41].

Assessing Care of Vulnerable Elders project — Another expert panel has identified quality indicators for appropriate medication use as part of the Assessing Care of Vulnerable Elders (ACOVE) project [78,79]. These indicators begin with practical suggestions on how to improve prescribing practices:

- Document the indication for a new drug therapy
- Educate patients on the benefits and risks associated with the use of a new therapy
- Maintain a current medication list
- Document response to therapy
- Periodically review the ongoing need for a drug therapy

In addition, these indicators specify drug therapies that either should not be prescribed for older adults or that warrant careful monitoring after they have been initiated ([table 4](#)).

UNDERUTILIZATION OF APPROPRIATE MEDICATION

Much attention has been paid to over-prescribing for older adults; under-prescribing appropriate medications is also of concern. Prescribing strategies that seek to simply limit the overall number of drugs prescribed to older adults in the name of improving quality of care may be seriously misdirected.

Clinicians may be better at avoiding over-prescribing of inappropriate drug therapies than at prescribing indicated drug therapies. As an example, one study of older adults (n = 372) in two managed care organizations found that 50 percent had not been prescribed some recommended therapy, while only 3 percent were prescribed medications classified as inappropriate [80]. Under- and overutilization of medications were equally prevalent in another study [81]. In a US Department of Veterans Affairs (VA) outpatient population, mean age 75 years (n = 196), inappropriate medications were documented for 65 percent and medication underuse for 64 percent; simultaneous under and overutilization occurred in 42 percent of patients.

START (Screening Tool to Alert doctors to the Right Treatment) is a set of validated criteria included within STOPP/START Version 3, developed by a consensus process involving experts in geriatric pharmacotherapy, aimed to identify potential prescribing omissions in older hospitalized patients [42].

It should also be recognized that determination of "under-prescribing" is based on guidelines that address individual disease entities, while most older adults have multiple conditions [23]. As an example, a patient with a myocardial infarction, history of diabetes, and elevated lipids would require a beta-blocker, angiotensin-converting enzyme (ACE) inhibitor, [aspirin](#), statin, and a hypoglycemic medication. Accordingly, many older adults need to take six or more essential medications. In this context, clinicians may make informed decisions to "under-prescribe" to foster compliance with essential medications, limit drug interactions, and prioritize health benefits for active treatment of serious conditions over preventive therapies or conditions that have less impact on quality of life.

Factors leading to unintended underutilization include clinicians not recognizing medication benefit in the older population, affordability, and dose availability.

Medication effectiveness — Studies of drug effectiveness specifically often exclude the older population due to concerns with comorbidities and adverse effects, causing difficulty in interpretation of study results. Therefore, the benefit of treatment for older adults, especially for preventive purposes, may not be established or may not be recognized by prescribing clinicians. As an example, in a study of statin use for secondary prevention in patients over age 66, the likelihood of being prescribed statin therapy declined 6.4 percent for every year of

age; overall, only 19 percent of patients in this high-risk population had been prescribed a statin [82].

Affordability — A prescription may be written but not filled, or filled and not taken regularly, due to an inability to pay for medications. This may be a particular problem in countries where there is no universal insurance coverage for drug therapy for older adults.

Enhanced drug coverage for older adults can be a powerful incentive to improve the use of beneficial therapies. A comparison of two groups of Medicare patients, as an example, found that statin use was 4.1 percent in patients without drug coverage and 27 percent in those with drug benefits [83]. Significant utilization differences between insured and uninsured patients were seen even for the use of inexpensive medications such as beta-blockers and nitrates.

Cost-related medical noncompliance affected almost 30 percent of disabled Medicare enrollees in 2004, and noncompliance rates were significantly higher for patients with multiple comorbidities [84].

Additional information on the affordability of medications can be found elsewhere in UpToDate. (See "[Patient education: Coping with high prescription drug prices in the United States \(Beyond the Basics\)](#)".)

Dose availability — Older individuals often require lower than usual doses of medications, especially at initiation. If medications are not readily available in prescribed doses, the need to split tablets may make it more difficult for patients to take beneficial drug therapy [85].

ADVERSE DRUG EVENTS

Adverse drug events (ADEs) are injuries that occur from use of a drug, including noxious responses, drug administration errors, and any other circumstances that lead to an injury. A number of factors in older adults contribute to their increased risk for developing ADEs, including frailty, coexisting medical problems, memory issues, and use of multiple prescribed and non-prescribed medications [86].

Types

Hospitalizations — Adverse drug reactions (ADRs) are responsible for between 3 to 10 percent of all hospitalizations among older patients [87-90]. Compared with younger adults, ADR-related hospitalizations in older adults are more common and are more likely to be preventable [88,91].

In the United States, between 2007 and 2009, an estimated 99,628 (95% CI 55,531-143,724) ADE-related hospitalizations occurred annually among individuals 65 years and older; two-thirds were due to unintentional overdoses [68]. Four medications or medication classes

were implicated in 67 percent of hospitalizations: [warfarin](#), insulins, oral antiplatelet agents, and oral hypoglycemic agents.

A 2017 systematic review and meta-analysis of 42 studies of hospitalizations among adults 60 years of age and older, conducted in 21 countries (the majority in Europe), found a mean prevalence of ADR-related hospitalizations of 8.7 percent (95% CI 7.6-9.8), with nonsteroidal antiinflammatory drugs (NSAIDs) being the most commonly implicated class of medications [92].

Prescribing cascades — Prescribing cascades occur when an adverse drug effect is misdiagnosed as a new medical condition, and treated with a potentially unnecessary drug [32,93]. Clinicians alert to the possibility of prescribing cascades can utilize tools to identify them [94]. The patient is then at risk for developing additional ADEs related to the new and potentially unnecessary treatment ([table 5](#)). Older adults with chronic conditions and multiple drug therapies are at particular risk for prescribing cascades. A consensus list of 65 potentially inappropriate prescribing cascades was created by an international group of experts from eight countries using a modified Delphi process [95]. This list complements existing tools designed to optimize prescribing by identifying potentially inappropriate drug combinations for further review.

Drug-induced symptoms in an older person can be easily misinterpreted as indicating a new disease or attributed to the aging process itself rather than the drug therapy. This misinterpretation is particularly likely when the drug-induced symptoms are indistinguishable from illnesses that are common in older persons. Selected examples of prescribing cascades are described below.

- One of the best recognized examples of a prescribing cascade relates to the initiation of antiparkinson therapy for symptoms arising from use of drugs such as antipsychotics [96-98] or [metoclopramide](#) [99]. The antiparkinson drugs can then lead to new symptoms, including orthostatic hypotension and delirium. (See "[Drug-induced parkinsonism](#)".)

In a case-control study of 3512 Medicaid patients (age 65 to 99 years), patients who had received an antipsychotic medication in the preceding 90 days were 5.4 times more likely to be prescribed antiparkinson therapy than patients who had not received an antipsychotic (95% CI 4.8-6.1) [96].

- Some prescribing cascades may be less obvious, especially for drugs whose adverse events are not as commonly recognized. As an example, cholinesterase inhibitors (eg, [donepezil](#), [rivastigmine](#), and [galantamine](#)) are commonly used for the management of dementia symptoms in older adults. The adverse events associated with these drugs can be viewed as the reverse of those that might be expected with anticholinergic therapies. Accordingly, while anticholinergic therapies may cause constipation and urinary retention, cholinesterase inhibitors may cause diarrhea and urinary incontinence. A

prescribing cascade occurs when the prescription of a cholinesterase inhibitor is followed by a prescription for an anticholinergic therapy (eg, [oxybutynin](#)) to treat incontinence.

A retrospective cohort study in older adults in Canada (n = 44,884) found that the risk of treatment with an anticholinergic medication for urinary incontinence was greater for patients who had received a cholinesterase inhibitor (adjusted hazard ratio 1.53; 95% CI 1.39-1.72) [[100](#)]. This study suggests that clinicians should consider the possible contributing role of cholinesterase inhibitors in new-onset or worsening urinary incontinence.

- Patients treated with calcium channel blockers can develop peripheral edema due to fluid redistribution, which may be interpreted as a new condition and treated with a diuretic.

In a retrospective cohort study of over 40,000 older adults with hypertension, 1.4 percent of patients prescribed a calcium channel blocker were subsequently prescribed a loop diuretic within 90 days compared with 0.7 percent of patients prescribed a different antihypertensive [[101](#)]. Since the peripheral edema from calcium channel blockers is due to fluid redistribution rather than fluid overload, treatment with a diuretic is not indicated and can lead to adverse outcomes including overdiuresis, falls, incontinence, acute kidney injury, and electrolyte imbalances.

Drug-drug interactions — Older adults are particularly vulnerable to drug-drug interactions because they often have multiple chronic medical conditions requiring multiple drug therapies. The risk of an adverse event due to drug-drug interactions is substantially increased when multiple drugs are taken [[102-106](#)]. As an example, the risk of bleeding with [warfarin](#) therapy is increased with coadministration of selective and nonselective NSAIDs, selective serotonin reuptake inhibitors, [omeprazole](#), lipid-lowering agents, [amiodarone](#), and [fluorouracil](#) [[102](#)].

A case control study from Canada evaluated hospitalizations for drug-related toxicity in a population of older patients who had received one of three drug therapies: [glyburide](#), [digoxin](#), or angiotensin-converting enzyme (ACE) inhibitor [[106](#)]. Hospitalization for hypoglycemia was six times more likely in patients who had received co-trimoxazole. Digoxin toxicity was 12 times more likely for patients who had been started on [clarithromycin](#). Hyperkalemia was 20 times more likely for patients who were treated with a potassium-sparing diuretic.

Care must be taken when prescribing any medication, especially for the older individual, to review existing medications and consider potential drug interactions.

Dose-related adverse drug events — ADEs are often dose-related. Examples include:

- A case-control study from the 1980s related risk of hip fracture in a Medicaid population with use and dose of psychotropic drugs [107]. A dose-related effect was seen for use of long half-life hypnotic-anxiolytics, tricyclic antidepressants, or neuroleptic therapy, and hospitalization for hip fracture.
- A study of people 65 years and older in Quebec (n >250,000) found that more than a quarter (27.6 percent) were dispensed at least one prescription for a benzodiazepine [108]. The risk of injury was dose-related for some benzodiazepines (oxazepam, flurazepam, and chlorthalidone), though not for alprazolam.
- Dose of benzodiazepine, but not elimination half-life, was related to risk for hip fracture in a case-control study of adults aged 55 years and older from the Netherlands [6].

Populations at higher risk — Certain patient groups are at higher risk for ADEs

Renal impairment — A common cause of dose-related adverse events in older adults is failure to properly adjust doses for renal insufficiency. Renal impairment becomes more common with advancing age. For patients with stable kidney function, creatinine clearance can be estimated according to published formulas which factor age into the calculation (calculator 1). Because of decreased muscle mass in older adults, however, serum creatinine levels may not adequately reflect renal function; many older patients with a normal creatinine nonetheless have modestly impaired kidney function. In one study, 40 percent of almost 10,000 older adults living in long-term care were found to have renal insufficiency [109]. In a community population over age 65 in France, the prevalence of renal insufficiency (estimated glomerular filtration rate [GFR] <60 mL/min/1.73 m²) was 13.7 percent using the MDRD equation and 36.9 percent using the Cockcroft-Gault formula [110]. (See "Assessment of kidney function".)

Dosing guidelines for decreased creatinine clearance are available to calculate dose adjustments for medications that are cleared through the kidney [111]. The list of medications is long and includes many antibiotics. In a community population, 52 percent of adults over age 65 with mild renal insufficiency were taking medications that required dose adjustment for low GFR; antihypertensives, fibrates, sedative/hypnotic, and anxiolytic medications accounted for most of these drugs [110]. The drug database available through UpToDate includes appropriate dose adjustments for kidney function and for older adults, and can be accessed by searching on any individual drug. As a general rule, the initial dose for starting medications in older adults should be significantly reduced, and titrated up as tolerated by monitoring side effects or drug levels.

Decision aids have been moderately effective in decreasing the percentage of in-hospital prescriptions written with inappropriate adjustments for renal status (46 to 33 percent) [112].

Residents in long-term care settings — Long-term care residents are at a particularly high risk for developing adverse events [113]. The average United States nursing home resident uses seven to eight different medications each month, and about one-third of residents have monthly drug regimens of nine or more medications [114].

A study of ADEs in two large academic long-term care facilities in the United States and Canada found 815 ADEs occurring during 8336 resident months [113]. The overall rate of ADEs was 9.8 per 100 resident-months; 42 percent of the ADEs were deemed preventable. Of the more serious adverse events, 61 percent were deemed preventable. The more serious the adverse event, the more likely it was to be considered potentially preventable. These rates were approximately four-times higher than had been previously reported [115] but may reflect the better documentation of ADEs at these institutions.

Preventable ADEs were most frequently associated with atypical antipsychotics and warfarin therapy (table 6). Neuropsychiatric events (confusion, oversedation, delirium), hemorrhagic events, and gastrointestinal events were the most frequent types of ADEs in the long-term care facilities studied (table 7). In a 12-month observational study of 490 long-term care residents taking warfarin in 25 nursing homes, there were 720 ADEs (625 minor, 82 serious, and 13 life-threatening); 57 percent of the serious events were considered preventable [116].

Antipsychotics — Antipsychotic medications, used for the management of the behavioral and psychological symptoms of dementia, are among the drugs most frequently associated with adverse events in long-term care facilities [113]. In particular, psychotropic medications are associated with an increased risk for falls. In one meta-analysis of patients age 60 or older, the odds ratio for any psychotropic use among patients who had one or more falls was 1.73 (95% CI 1.52-1.97) [117].

There is limited evidence to support the efficacy of these agents for management of behavioral and psychological symptoms in older adults. (See "[Management of neuropsychiatric symptoms of dementia](#)", section on 'Antipsychotic drugs' and "[Second-generation and other antipsychotic medications: Pharmacology, administration, and side effects](#)".)

Nonetheless, use of antipsychotic medications in long-term care facilities is widespread. A study of 19,780 older adults with no history of major psychosis prior to long-term care admission found that antipsychotic therapy was prescribed for 17 percent within 100 days of their long-term care admission and for 24 percent within one year [118]. A study of 485 nursing homes in Canada found that there was about a threefold variation in antipsychotic prescribing, not related to clinical factors, between high- and low-prescribing facilities [119].

A public health advisory warning issued from the US Food and Drug Administration (FDA) warns of fatal adverse events in patients with dementia treated with atypical antipsychotic therapy [120-122]. Data from 17 trials of older adult patients with dementia have shown that those treated with atypical antipsychotic therapy were 1.6 to 1.7 times more likely to die than those given placebo therapy. Similar concerns have been raised for [haloperidol](#) and other conventional antipsychotics [123,124]. A retrospective comparison of patients with dementia who were newly treated with atypical antipsychotics, compared with no antipsychotics, found an increased risk of death at 30 and 180 days for the treated group (at 30 days, adjusted hazard ratio [HR] 1.55, 95% CI 1.15-2.07) [125]. Mortality was further increased, again by a factor of 1.55, for patients receiving conventional antipsychotics compared with atypical antipsychotics. These data point to the need to rethink the role of these therapies in clinical practice. (See ["Management of neuropsychiatric symptoms of dementia", section on 'Severe or refractory symptoms'](#).)

Predicting adverse drug reactions — A tool has been developed to identify older adult patients at increased risk for an ADR in hospital [126]. The tool, based on logistic regression analysis from a group of Italian patients mean age 78, and validated in a separate European cohort, found that the number of drugs prescribed and prior history of an ADR were the strongest predictors for a subsequent ADR. Compared with those receiving five or fewer medications, the risk of ADR was approximately doubled (odds ratio [OR] 1.9, 95% CI 1.35-2.68) for those prescribed five to seven medications, and was fourfold (OR 4.07, CI 2.93-5.65) for those receiving eight or more medications. Other variables incorporated in this tool are the presence of four or more comorbid conditions, heart failure, liver disease, or kidney failure.

Preventing adverse drug events — The occurrence of preventable ADEs is a significant concern. Inappropriate ordering and inadequate monitoring are the most common errors in preventable adverse drug events. Errors in transcription, dispensing, and administration are less commonly identified [113].

Medications that are commonly implicated in preventable ADEs are not generally those identified by widely utilized "bad drug" lists. "Good drugs" prescribed in an inappropriate manner may be far more problematic. When drugs do cause problems, it is often because they are prescribed, dosed, or monitored inappropriately.

Prevention of ADEs in the hospital setting is discussed separately. (See ["Prevention of adverse drug events in hospitals"](#).)

Long-term care — Enhanced surveillance and reporting systems for ADEs occurring in the nursing home setting are needed. Computerized order entry in the hospital setting has been shown to reduce serious medication errors [127]. A computer-based decision aid

reduced in-hospital inappropriate dosing of psychotropic medications for geriatric inpatients [128]. A randomized trial of computerized order entry with clinical decision support in 29 resident care units at two long-term care homes in Canada did not affect the rate of ADEs [129].

Community care — Patient errors in medication adherence are a significant contributor to ADEs for older patients living in the community, accounting for 21 percent of preventable ADEs in a large ambulatory Medicare population [130]. Patient errors occurred more frequently in patients who were regularly taking three or more medications, compared with those taking two or fewer [131].

Practical recommendations to reduce medical errors in the community have been proposed [132-137]:

- Maintain an accurate list of all medications that a patient is currently using. This list should include the drug name (generic and brand), dose, frequency, route, and indication.
- Advise periodic "brown-bag check-ups." Instruct patients to bring all pill bottles to each medical visit; bottles should be checked against the medication list.
- Patients should be made aware of potential drug confusions: sound-alike names, look-alike pills, and combination medications.
- Patients should be informed of both generic and brand names, including spelling, as well as the reasons for taking their medications. This may prevent unnecessary confusion when drugs are inconsistently labeled. As an example, a patient may be unaware that [digoxin](#) (generic) and Lanoxin (brand) are the same therapy.
- Medication organizers that are filled by the patient, family member, or caregiver can facilitate compliance with drug regimens. Blister packs for individual drugs, prepared by the pharmacist, can also be helpful in ensuring that patients take their medications correctly [136].
- Community pharmacists are an important resource and can play a key role in working with older adults to reduce medication errors.

Transitions in care settings — Transitions in care, between hospital and nursing home or institutional setting and home, are a common source of medication errors and confusions:

- One Canadian multisite study found that 23 percent of 328 older adults experienced an ADE after discharge home from the hospital; half of these ADEs were considered preventable [132].

- Changes in medication (different dose, discontinued therapies, additional therapies) were identified in 45 of 50 patients discharged from a geriatric ward in the United Kingdom within 6 to 14 days of discharge [133]. Of particular concern is discharge of older patients with new prescriptions for benzodiazepines that were initiated in the hospital, leading to unplanned chronic benzodiazepine use [138].
- Attending clinicians from an academic medical center reported that they believed 89 percent of their discharged patients (n = 99) understood potential side effects of their medications; 58 percent of those discharged patients reported that they understood this information [139].
- ADEs attributed to medication changes occurred in 20 percent of patients on transfer from hospital to a nursing home, occurring most commonly for patients being readmitted to the nursing home (12 of 14 events) [140].
- Frail older people are often found to be on unnecessary drug regimens at the time of hospital discharge. Among 384 older veterans, 44 percent were found to have at least one unnecessary drug therapy at the time of discharge [141]. Factors contributing to this include multiple prescribers, "routine" medications for hospitalization such as antacids or stool softeners, and being on nine or more drug therapies.

Effort must be made to improve communication in "hand-offs" of patient care during transitions in care setting. This is particularly true when the physician responsible for the patient in the hospital is not the same as the physician providing the patient's longitudinal care. Accurate medication lists, direct communications between providers, and a thorough review of all medications at the time of care transition for appropriateness and intended duration of treatment, are steps that should be taken to avoid ADEs. Whenever possible, the number of prescribing physicians for an individual patient should be limited, as the number of prescribing physicians is an independent risk factor for ADEs [142]. Safe and effective hospital discharge principles are discussed separately. (See "[Hospital discharge and readmission](#)".)

A STEPWISE APPROACH TO PRESCRIBING

Presented below is one systematic approach to improving prescribing practices when managing older adults. Other systematic approaches have been described incorporating similar elements [143]. Regardless of the sequence of steps, what is essential in prescribing is to continually reappraise the patient's medication regimen in light of his or her current clinical status, goals of care, and the potential risks/benefits of each medication.

A concept of "time to benefit" (TTB) in relation to drug prescribing for older patients with multiple morbidities can be applied to therapeutic decisions [144]. TTB, defined as the time to

significant benefit observed in trials of people treated with a drug compared with controls, can be estimated from data from randomized controlled trials. Such information, not routinely available, may in the future help guide decision-making for specific drug prescribing in individual patients.

Review current drug therapy — Periodic evaluation of a patient's drug regimen is an essential component of medical care for an older person. Such a review may indicate the need for changes to prescribed drug therapy. These changes may include discontinuing a therapy prescribed for an indication that no longer exists, substituting a therapy with a potentially safer agent, changing a drug dose, or adding a new medication ([table 8](#)). A medication review should consider whether a change in patient status (eg, renal or liver function) might necessitate dosing adjustment, the potential for drug-drug interaction, whether patient symptoms might reflect a drug side effect, or whether the regimen could be simplified [145]. Medication reviews are often not done in a systematic manner. A reasonable approach could be having a patient meet with a pharmacist within a few weeks of starting a new medication.

In addition to routine review of therapy, review of drug therapy is indicated when patients present with an injury or illness that might have been an adverse result of a prescribed medication. As an example, one study reviewed data for a sample of 168,000 Medicare patients seen for medical care with a fracture of a hip, shoulder, or wrist [146]. In the four months prior to presentation, three-quarters of the patients had been taking a nonopioid drug associated with increased fracture risk (eg, sedative, atypical antipsychotic, or antihypertensive). In the four months after the fracture, such drugs were discontinued for 7 percent but were newly prescribed for another 7 percent.

In a survey of Medicare beneficiaries, more than 30 percent of patients reported they had not talked with their doctor about their different medications in the previous 12 months [147]. Ideally, the clinician should ask the patient to bring to the visit all of the bottles of pills that they are using. Patients may not consider over-the-counter products, ointments, vitamins, ophthalmic preparations, or herbal medicines to be drug therapies and need to be specifically told to bring these to the visit.

Unintended medication discrepancies, particularly likely to occur at the time of hospital admissions, are a common source for medication errors. As an example, one study evaluated 151 patients (average age 77 years) admitted to general internal medicine clinical teaching units and found discrepancies in more than half between admission medication orders and the patient's usual drug therapies as identified by a medication history interview [148]. Most discrepancies involved unintentional omission of a maintenance medication and more than a third of these discrepancies had the potential to cause moderate harm.

Discontinue unnecessary therapy — Clinicians are often reluctant to stop medications, especially if they did not initiate the treatment and the patient seems to be tolerating the

therapy. Sometimes, this exposes the patient to the risks for an adverse event with limited therapeutic benefit. A common example is the use of [digoxin](#) in older adults, often prescribed for indications that have not been well-documented. Renal impairment or temporary dehydration may predispose older adults to digoxin toxicity [149]. Although digoxin therapy can be safely discontinued in selected nursing home residents, it is important to recognize that discontinuation in patients with impaired systolic function can have a detrimental effect [150]. (See "[Overview of the management of heart failure with reduced ejection fraction in adults](#)".)

The decision to discontinue medication is determined in part by the goals of care for that patient and the risks of adverse effects for that patient. Targets for treatment, based on outcomes evidence from studies in younger patients, may not be appropriate for older adults [33]; thus clinical guidelines not targeted to older patients may foster overly aggressive goals for management of hypertension or diabetes in the older adult population.

One approach to assessing whether a drug is truly necessary for a given patient is presented in an algorithm ([algorithm 1](#)) [151]. In a feasibility study performed in a cohort of 70 community-dwelling patients seen for geriatric assessment, implementation of this algorithm led to recommendations to discontinue 58 percent of the medications they had been taking. Eighty-one percent of these medications were discontinued, 2 percent were restarted, and no significant adverse events were attributable to discontinuation over 13-month follow-up.

Some preventive and other therapies may no longer be beneficial to patients with short life expectancies [35]. The appropriateness of these therapies should be reconsidered when other medical conditions develop that impact a patient's long-term prognosis, unless the therapies are thought to increase comfort.

There are limited studies about how best to withdraw medications [34]. It is reasonable to gradually taper off most medications to minimize withdrawal reactions and to allow symptom monitoring, unless dangerous signs or symptoms indicate a need for abrupt medication withdrawal. Certain common drugs require tapering, including beta blockers, opioids, barbiturates, [clonidine](#), [gabapentin](#), and antidepressants.

Deprescribing is discussed in more detail separately. (See "[Deprescribing](#)".)

Consider adverse drug events for any new symptom — Before adding a new therapy to the patient's drug regimen, clinicians should carefully consider whether the development of a new medical condition could be the presentation of an atypical ADE to an existing drug therapy. Many prescribing cascade scenarios have been identified ([table 5](#)). (See '[Prescribing cascades](#)' above.)

Consider nonpharmacologic approaches — Some conditions in older adults may be amenable to lifestyle modification in lieu of pharmacotherapy. The Trial of Nonpharmacologic

Interventions in the Elderly (TONE) demonstrated that weight loss and reduced sodium intake could allow discontinuation of antihypertensive medication in about 40 percent of the intervention group [152,153]. (See "[Treatment of hypertension in older adults, particularly isolated systolic hypertension](#)", section on 'Lifestyle modifications'.)

Care in the use of common drugs — Some commonly prescribed drugs may result in increased toxicity in older adults. As an example, numerous studies have documented adverse events associated with nonsteroidal antiinflammatory drug (NSAID) use, including gastrointestinal bleeding [154], renal impairment [155], and heart failure in this population [156]. NSAIDs should be used cautiously in older adults and generally for a limited duration. (See "[Nonselective NSAIDs: Overview of adverse effects](#)".)

Reduce the dose — Many ADEs are dose-related. When prescribing drug therapies, it is important to use the minimal dose required to obtain clinical benefit. As an example, one study evaluated the relationship between prescribing of the newer atypical antipsychotic therapies (eg, [olanzapine](#), [risperidone](#), and [quetiapine](#)) and the development of parkinsonism in older adults [97]. Relative to those dispensed a low dose, those dispensed a high dose were more than twice as likely to develop parkinsonism (HR 2.07, 95% CI 1.42-3.02). As another example, one case-control study in patients over age 70 who received thyroid supplementation identified a correlation between risk of fracture and dose of [levothyroxine](#), indicating the importance of testing for thyroid levels in this population and adjusting the dose accordingly [157].

Simplify the dosing schedule — When multiple medications are required, greater regimen complexity will increase the likelihood of poor compliance or confusion with dosing. Older adults, and particularly those with low health literacy or cognitive impairment, are not able to efficiently consolidate prescription regimens to optimize a dosing schedule [158]. The Institute of Medicine has proposed a standardized schedule for specifying medication dosing (morning, noon, evening, bedtime), recognizing that 90 percent of prescriptions are taken four or fewer times daily [159].

Simplifying the medication dosing schedule, when possible, is also important in the long-term care setting where nursing staff and time requirements for medication administration are substantial. A study illustrated that within a seven-hour shift, on a 20-bed unit, with two scheduled periods of medication administration, the process of administering medications to the residents accounted for a third of the nursing time [160]. This makes the nurse less available for other important patient care tasks.

Prescribe beneficial therapy — The fewer-the-better approach to drug therapy in older adults is often not the best response to optimizing drug regimens. Avoiding medications with known benefits to minimize the number of drugs prescribed is inappropriate. Patients must

be informed about the reason to initiate a new medication and what the expected benefits are.

CHALLENGES FOR CAREGIVERS MANAGING MEDICATIONS

A major challenge faced by informal caregivers is managing medications for an older person. In a United States survey, almost 80 percent of such caregivers managed medication regimens, and many found this stressful. The following are four ways that providers can support caregivers in their medication management [161]:

- **Information sharing** – Provide caregivers with access to a provider to ask questions about medications and to reinforce the importance of talking with providers before making changes to medications.
 - **Scheduling logistics and simplification** – Keep the regimens as simple as possible. Focus on medications that support goals of care. Simplify dosing schedules. Provide simple, clear instructions.
 - **Safety issues** – Ensure that the caregivers understand what the medication is, what it is for, and how it needs to be given. Medications need to be kept out of reach of those with cognitive impairment.
 - **Polypharmacy** – With polypharmacy comes the risk of medication errors and the increased risk of adverse events. Review medications for opportunities to deprescribe those that provide little benefit.
-

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "[Patient education: Taking medicines when you're older \(The Basics\)](#)" and "[Patient education: Side effects from medicines \(The Basics\)](#)" and "[Patient education: Medication safety \(The Basics\)](#)")

SUMMARY AND RECOMMENDATIONS

- **Recognizing adverse drug events** – The possibility of an adverse drug event (ADE) should always be borne in mind when evaluating an older adult; any new symptom should be considered drug-related until proven otherwise. Pharmacokinetic changes lead to increased plasma drug concentrations and pharmacodynamic changes lead to increased drug sensitivity in older adults. (See '[Introduction](#)' above.)
- **Role of herbal and dietary supplements** – Clinicians must be alert to the use of herbal and dietary supplements by older patients, who may not volunteer this information and are prone to drug-drug interactions related to these supplements. (See '[Herbal and dietary supplements](#)' above.)
- **Potentially inappropriate medications** – Various criteria sets exist identifying medications that should not be prescribed, or should be prescribed with great caution, in older adults. Clinicians need to consider each patient's individual situation, and they should use their best clinical judgment rather than strictly adhere to prescribing guidelines when making prescribing decisions. (See '[Potentially inappropriate medications](#)' above.)
- **Underutilization of appropriate medication** – Clinicians also under-prescribe medications, such as statins, that could provide benefit for older adults. Clinicians may be better at avoiding overprescribing of inappropriate drug therapies than at prescribing indicated drug therapies. Patient financial constraints and unavailability of prescribed doses may contribute to medication underutilization. (See '[Underutilization of appropriate medication](#)' above.)
- **Risk of adverse drug events**
 - **Older adults** – ADEs result in four times as many hospitalizations in older, compared with younger, adults. Prescribing cascades, drug-drug interactions, and inappropriate drug doses are causes of preventable ADEs. (See '[Adverse drug events](#)' above.)
 - **Residents in long-term care settings** – ADEs are a particular problem for nursing home residents; atypical antipsychotic medications and [warfarin](#) are the most common drugs involved in ADEs in this population. (See '[Residents in long-term care settings](#)' above.)
- **Stepwise approach to prescribing** – A stepwise approach to prescribing for older adults should include: periodic review of current drug therapy; discontinuing unnecessary medications; considering nonpharmacologic alternative strategies; considering safer

alternative medications; using the lowest possible effective dose; including all necessary beneficial medications. (See '[A stepwise approach to prescribing](#)' above.)

Use of UpToDate is subject to the [Terms of Use](#).

REFERENCES

1. Cho S, Lau SW, Tandon V, et al. Geriatric drug evaluation: where are we now and where should we be in the future? *Arch Intern Med* 2011; 171:937.
2. Rowe JW, Andres R, Tobin JD, et al. The effect of age on creatinine clearance in men: a cross-sectional and longitudinal study. *J Gerontol* 1976; 31:155.
3. Reidenberg MM, Levy M, Warner H, et al. Relationship between diazepam dose, plasma level, age, and central nervous system depression. *Clin Pharmacol Ther* 1978; 23:371.
4. Pomara N, Stanley B, Block R, et al. Adverse effects of single therapeutic doses of diazepam on performance in normal geriatric subjects: relationship to plasma concentrations. *Psychopharmacology (Berl)* 1984; 84:342.
5. Pomara N, Stanley B, Block R, et al. Increased sensitivity of the elderly to the central depressant effects of diazepam. *J Clin Psychiatry* 1985; 46:185.
6. Herings RM, Stricker BH, de Boer A, et al. Benzodiazepines and the risk of falling leading to femur fractures. Dosage more important than elimination half-life. *Arch Intern Med* 1995; 155:1801.
7. Scott JC, Stanski DR. Decreased fentanyl and alfentanil dose requirements with age. A simultaneous pharmacokinetic and pharmacodynamic evaluation. *J Pharmacol Exp Ther* 1987; 240:159.
8. Tan JL, Eastment JG, Poudel A, Hubbard RE. Age-Related Changes in Hepatic Function: An Update on Implications for Drug Therapy. *Drugs Aging* 2015; 32:999.
9. Qato DM, Wilder J, Schumm LP, et al. Changes in Prescription and Over-the-Counter Medication and Dietary Supplement Use Among Older Adults in the United States, 2005 vs 2011. *JAMA Intern Med* 2016; 176:473.
10. Saraf AA, Petersen AW, Simmons SF, et al. Medications associated with geriatric syndromes and their prevalence in older hospitalized adults discharged to skilled nursing facilities. *J Hosp Med* 2016; 11:694.
11. Kaufman DW, Kelly JP, Rosenberg L, et al. Recent patterns of medication use in the ambulatory adult population of the United States: the Slone survey. *JAMA* 2002; 287:337.
12. Nahin RL, Pecha M, Welmerink DB, et al. Concomitant use of prescription drugs and dietary supplements in ambulatory elderly people. *J Am Geriatr Soc* 2009; 57:1197.
13. Eisenberg DM, Kessler RC, Foster C, et al. Unconventional medicine in the United States. Prevalence, costs, and patterns of use. *N Engl J Med* 1993; 328:246.

14. Fugh-Berman A. Herb-drug interactions. *Lancet* 2000; 355:134.
15. Wold RS, Lopez ST, Yau CL, et al. Increasing trends in elderly persons' use of nonvitamin, nonmineral dietary supplements and concurrent use of medications. *J Am Diet Assoc* 2005; 105:54.
16. Morris CA, Avorn J. Internet marketing of herbal products. *JAMA* 2003; 290:1505.
17. Spinewine A, Schmader KE, Barber N, et al. Appropriate prescribing in elderly people: how well can it be measured and optimised? *Lancet* 2007; 370:173.
18. Sacarny A, Barnett ML, Le J, et al. Effect of Peer Comparison Letters for High-Volume Primary Care Prescribers of Quetiapine in Older and Disabled Adults: A Randomized Clinical Trial. *JAMA Psychiatry* 2018; 75:1003.
19. Alldred DP, Kennedy MC, Hughes C, et al. Interventions to optimise prescribing for older people in care homes. *Cochrane Database Syst Rev* 2016; 2:CD009095.
20. Wolfstadt JI, Gurwitz JH, Field TS, et al. The effect of computerized physician order entry with clinical decision support on the rates of adverse drug events: a systematic review. *J Gen Intern Med* 2008; 23:451.
21. Ferner RE, Aronson JK. Communicating information about drug safety. *BMJ* 2006; 333:143.
22. Duerden M, Avery T, Payne R. Polypharmacy and medicines optimisation: Making it safe and sound. The King's Fund 2013. Available at: kingsfund.org.uk/publications/polypharmacy-and-medicines-optimisation (Accessed on April 21, 2021).
23. Tinetti ME, Bogardus ST Jr, Agostini JV. Potential pitfalls of disease-specific guidelines for patients with multiple conditions. *N Engl J Med* 2004; 351:2870.
24. Nightingale G, Hajjar E, Swartz K, et al. Evaluation of a pharmacist-led medication assessment used to identify prevalence of and associations with polypharmacy and potentially inappropriate medication use among ambulatory senior adults with cancer. *J Clin Oncol* 2015; 33:1453.
25. Field TS, Gurwitz JH, Avorn J, et al. Risk factors for adverse drug events among nursing home residents. *Arch Intern Med* 2001; 161:1629.
26. Wimmer BC, Cross AJ, Jekanovic N, et al. Clinical Outcomes Associated with Medication Regimen Complexity in Older People: A Systematic Review. *J Am Geriatr Soc* 2017; 65:747.
27. Lu WH, Wen YW, Chen LK, Hsiao FY. Effect of polypharmacy, potentially inappropriate medications and anticholinergic burden on clinical outcomes: a retrospective cohort study. *CMAJ* 2015; 187:E130.
28. Rawle MJ, Cooper R, Kuh D, Richards M. Associations Between Polypharmacy and Cognitive and Physical Capability: A British Birth Cohort Study. *J Am Geriatr Soc* 2018; 66:916.

29. Fried TR, O'Leary J, Towle V, et al. Health outcomes associated with polypharmacy in community-dwelling older adults: a systematic review. *J Am Geriatr Soc* 2014; 62:2261.
30. Weng MC, Tsai CF, Sheu KL, et al. The impact of number of drugs prescribed on the risk of potentially inappropriate medication among outpatient older adults with chronic diseases. *QJM* 2013; 106:1009.
31. Lai SW, Liao KF, Liao CC, et al. Polypharmacy correlates with increased risk for hip fracture in the elderly: a population-based study. *Medicine (Baltimore)* 2010; 89:295.
32. Rochon PA, Gurwitz JH. Optimising drug treatment for elderly people: the prescribing cascade. *BMJ* 1997; 315:1096.
33. Boyd CM, Darer J, Boult C, et al. Clinical practice guidelines and quality of care for older patients with multiple comorbid diseases: implications for pay for performance. *JAMA* 2005; 294:716.
34. Steinman MA, Hanlon JT. Managing medications in clinically complex elders: "There's got to be a happy medium". *JAMA* 2010; 304:1592.
35. Holmes HM, Hayley DC, Alexander GC, Sachs GA. Reconsidering medication appropriateness for patients late in life. *Arch Intern Med* 2006; 166:605.
36. Mitchell SL, Teno JM, Kiely DK, et al. The clinical course of advanced dementia. *N Engl J Med* 2009; 361:1529.
37. Givens JL, Jones RN, Shaffer ML, et al. Survival and comfort after treatment of pneumonia in advanced dementia. *Arch Intern Med* 2010; 170:1102.
38. McLeod PJ, Huang AR, Tamblyn RM, Gayton DC. Defining inappropriate practices in prescribing for elderly people: a national consensus panel. *CMAJ* 1997; 156:385.
39. Fick DM, Cooper JW, Wade WE, et al. Updating the Beers criteria for potentially inappropriate medication use in older adults: results of a US consensus panel of experts. *Arch Intern Med* 2003; 163:2716.
40. Zhan C, Sangl J, Bierman AS, et al. Potentially inappropriate medication use in the community-dwelling elderly: findings from the 1996 Medical Expenditure Panel Survey. *JAMA* 2001; 286:2823.
41. Hanlon JT, Schmader KE, Boult C, et al. Use of inappropriate prescription drugs by older people. *J Am Geriatr Soc* 2002; 50:26.
42. O'Mahony D, Cherubini A, Guiteras AR, et al. STOPP/START criteria for potentially inappropriate prescribing in older people: version 3. *Eur Geriatr Med* 2023; 14:625.
43. Pazan F, Weiss C, Wehling M, FORTA. The EURO-FORTA (Fit fOR The Aged) List: International Consensus Validation of a Clinical Tool for Improved Drug Treatment in Older People. *Drugs Aging* 2018; 35:61.
44. Hilmer SN, Mager DE, Simonsick EM, et al. A drug burden index to define the functional

- burden of medications in older people. *Arch Intern Med* 2007; 167:781.
45. Hilmer SN, Mager DE, Simonsick EM, et al. Drug burden index score and functional decline in older people. *Am J Med* 2009; 122:1142.
 46. Hampton LM, Daubresse M, Chang HY, et al. Emergency department visits by adults for psychiatric medication adverse events. *JAMA Psychiatry* 2014; 71:1006.
 47. Wilson NM, Hilmer SN, March LM, et al. Associations between drug burden index and falls in older people in residential aged care. *J Am Geriatr Soc* 2011; 59:875.
 48. Tian F, Chen Z, Zeng Y, et al. Prevalence of Use of Potentially Inappropriate Medications Among Older Adults Worldwide: A Systematic Review and Meta-Analysis. *JAMA Netw Open* 2023; 6:e2326910.
 49. Sura SD, Carnahan RM, Chen H, Aparasu RR. Prevalence and determinants of anticholinergic medication use in elderly dementia patients. *Drugs Aging* 2013; 30:837.
 50. Paul KJ, Walker RL, Dublin S. Anticholinergic medications and risk of community-acquired pneumonia in elderly adults: a population-based case-control study. *J Am Geriatr Soc* 2015; 63:476.
 51. Carrière I, Fourrier-Reglat A, Dartigues JF, et al. Drugs with anticholinergic properties, cognitive decline, and dementia in an elderly general population: The 3-city study. *Arch Intern Med* 2009; 169:1317.
 52. Gray SL, Anderson ML, Dublin S, et al. Cumulative use of strong anticholinergics and incident dementia: a prospective cohort study. *JAMA Intern Med* 2015; 175:401.
 53. Fox C, Richardson K, Maidment ID, et al. Anticholinergic medication use and cognitive impairment in the older population: the medical research council cognitive function and ageing study. *J Am Geriatr Soc* 2011; 59:1477.
 54. Chew ML, Mulsant BH, Pollock BG, et al. Anticholinergic activity of 107 medications commonly used by older adults. *J Am Geriatr Soc* 2008; 56:1333.
 55. Salahudeen MS, Hilmer SN, Nishtala PS. Comparison of anticholinergic risk scales and associations with adverse health outcomes in older people. *J Am Geriatr Soc* 2015; 63:85.
 56. Beers MH, Ouslander JG, Rollinger I, et al. Explicit criteria for determining inappropriate medication use in nursing home residents. UCLA Division of Geriatric Medicine. *Arch Intern Med* 1991; 151:1825.
 57. <https://agsjournals.onlinelibrary.wiley.com/doi/10.1111/jgs.18372> (Accessed on June 06, 2023).
 58. Steinman MA, Beizer JL, DuBeau CE, et al. How to Use the American Geriatrics Society 2015 Beers Criteria-A Guide for Patients, Clinicians, Health Systems, and Payors. *J Am Geriatr Soc* 2015; 63:e1.
 59. American Geriatrics Society Beers Criteria® Alternatives Panel, Steinman MA. Alternative Treatments to Selected Medications in the 2023 American Geriatrics Society Beers

Criteria®. J Am Geriatr Soc 2025; 73:2657.

60. Lane CJ, Bronskill SE, Sykora K, et al. Potentially inappropriate prescribing in Ontario community-dwelling older adults and nursing home residents. J Am Geriatr Soc 2004; 52:861.
61. Fialová D, Topinková E, Gambassi G, et al. Potentially inappropriate medication use among elderly home care patients in Europe. JAMA 2005; 293:1348.
62. Davidoff AJ, Miller GE, Sarpong EM, et al. Prevalence of potentially inappropriate medication use in older adults using the 2012 Beers criteria. J Am Geriatr Soc 2015; 63:486.
63. Yang AW, Leng M, Ly DP, et al. Prescribing Patterns of Potentially Inappropriate CNS-Active Medications in Older Adults. JAMA 2026; 335:544.
64. Rochon PA, Lane CJ, Bronskill SE, et al. Potentially inappropriate prescribing in Canada relative to the US. Drugs Aging 2004; 21:939.
65. Lau DT, Kasper JD, Potter DE, et al. Hospitalization and death associated with potentially inappropriate medication prescriptions among elderly nursing home residents. Arch Intern Med 2005; 165:68.
66. Jano E, Aparasu RR. Healthcare outcomes associated with beers' criteria: a systematic review. Ann Pharmacother 2007; 41:438.
67. Budnitz DS, Shehab N, Kegler SR, Richards CL. Medication use leading to emergency department visits for adverse drug events in older adults. Ann Intern Med 2007; 147:755.
68. Budnitz DS, Lovegrove MC, Shehab N, Richards CL. Emergency hospitalizations for adverse drug events in older Americans. N Engl J Med 2011; 365:2002.
69. Gallagher P, O'Mahony D. STOPP (Screening Tool of Older Persons' potentially inappropriate Prescriptions): application to acutely ill elderly patients and comparison with Beers' criteria. Age Ageing 2008; 37:673.
70. Gallagher P, Ryan C, Byrne S, et al. STOPP (Screening Tool of Older Person's Prescriptions) and START (Screening Tool to Alert doctors to Right Treatment). Consensus validation. Int J Clin Pharmacol Ther 2008; 46:72.
71. O'Mahony D, O'Sullivan D, Byrne S, et al. STOPP/START criteria for potentially inappropriate prescribing in older people: version 2. Age Ageing 2015; 44:213.
72. O'Connor MN, O'Sullivan D, Gallagher PF, et al. Prevention of Hospital-Acquired Adverse Drug Reactions in Older People Using Screening Tool of Older Persons' Prescriptions and Screening Tool to Alert to Right Treatment Criteria: A Cluster Randomized Controlled Trial. J Am Geriatr Soc 2016; 64:1558.
73. Farhat A, Al-Hajje A, Lang PO, Csajka C. Impact of Pharmaceutical Interventions with STOPP/START and PIM-Check in Older Hospitalized Patients: A Randomized Controlled Trial. Drugs Aging 2022; 39:899.

74. Wehling M. Multimorbidity and polypharmacy: how to reduce the harmful drug load and yet add needed drugs in the elderly? Proposal of a new drug classification: fit for the aged. *J Am Geriatr Soc* 2009; 57:560.
75. Kuhn-Thiel AM, Weiß C, Wehling M, FORTA authors/expert panel members. Consensus validation of the FORTA (Fit FOR The Aged) List: a clinical tool for increasing the appropriateness of pharmacotherapy in the elderly. *Drugs Aging* 2014; 31:131.
76. Skains RM, Koehl JL, Aldeen A, et al. Geriatric Emergency Medication Safety Recommendations (GEMS-Rx): Modified Delphi Development of a High-Risk Prescription List for Older Emergency Department Patients. *Ann Emerg Med* 2024; 84:274.
77. Iscoe MS, Sangal RB, Hwang U, et al. High-Risk Medication Prescriptions Among Older Adults Discharged from the Emergency Department. *JAMA Intern Med* 2026; 186:483.
78. Shekelle PG, MacLean CH, Morton SC, Wenger NS. Acove quality indicators. *Ann Intern Med* 2001; 135:653.
79. Shrank WH, Polinski JM, Avorn J. Quality indicators for medication use in vulnerable elders. *J Am Geriatr Soc* 2007; 55 Suppl 2:S373.
80. Higashi T, Shekelle PG, Solomon DH, et al. The quality of pharmacologic care for vulnerable older patients. *Ann Intern Med* 2004; 140:714.
81. Steinman MA, Landefeld CS, Rosenthal GE, et al. Polypharmacy and prescribing quality in older people. *J Am Geriatr Soc* 2006; 54:1516.
82. Ko DT, Mamdani M, Alter DA. Lipid-lowering therapy with statins in high-risk elderly patients: the treatment-risk paradox. *JAMA* 2004; 291:1864.
83. Federman AD, Adams AS, Ross-Degnan D, et al. Supplemental insurance and use of effective cardiovascular drugs among elderly medicare beneficiaries with coronary heart disease. *JAMA* 2001; 286:1732.
84. Soumerai SB, Pierre-Jacques M, Zhang F, et al. Cost-related medication nonadherence among elderly and disabled medicare beneficiaries: a national survey 1 year before the medicare drug benefit. *Arch Intern Med* 2006; 166:1829.
85. Rochon PA, Anderson GM, Tu JV, et al. Age- and gender-related use of low-dose drug therapy: the need to manufacture low-dose therapy and evaluate the minimum effective dose. *J Am Geriatr Soc* 1999; 47:954.
86. Mallet L, Spinewine A, Huang A. The challenge of managing drug interactions in elderly people. *Lancet* 2007; 370:185.
87. Onder G, Pedone C, Landi F, et al. Adverse drug reactions as cause of hospital admissions: results from the Italian Group of Pharmacoepidemiology in the Elderly (GIFA). *J Am Geriatr Soc* 2002; 50:1962.
88. Beijer HJ, de Blaey CJ. Hospitalisations caused by adverse drug reactions (ADR): a meta-analysis of observational studies. *Pharm World Sci* 2002; 24:46.

89. Kongkaew C, Noyce PR, Ashcroft DM. Hospital admissions associated with adverse drug reactions: a systematic review of prospective observational studies. *Ann Pharmacother* 2008; 42:1017.
90. Alhawassi TM, Krass I, Bajorek BV, Pont LG. A systematic review of the prevalence and risk factors for adverse drug reactions in the elderly in the acute care setting. *Clin Interv Aging* 2014; 9:2079.
91. Budnitz DS, Pollock DA, Weidenbach KN, et al. National surveillance of emergency department visits for outpatient adverse drug events. *JAMA* 2006; 296:1858.
92. Oscanoa TJ, Lizaraso F, Carvajal A. Hospital admissions due to adverse drug reactions in the elderly. A meta-analysis. *Eur J Clin Pharmacol* 2017; 73:759.
93. Rochon PA, Gurwitz JH. The prescribing cascade revisited. *Lancet* 2017; 389:1778.
94. McCarthy LM, Savage R, Dalton K, et al. ThinkCascades: A Tool for Identifying Clinically Important Prescribing Cascades Affecting Older People. *Drugs Aging* 2022; 39:829.
95. Rochon PA, O'Mahony D, Cherubini A, et al. International expert panel's potentially inappropriate prescribing cascades (PIPC) list. *Eur Geriatr Med* 2025; 16:1573.
96. Avorn J, Bohn RL, Mogun H, et al. Neuroleptic drug exposure and treatment of parkinsonism in the elderly: a case-control study. *Am J Med* 1995; 99:48.
97. Rochon PA, Stukel TA, Sykora K, et al. Atypical antipsychotics and parkinsonism. *Arch Intern Med* 2005; 165:1882.
98. Stephen PJ, Williamson J. Drug-induced parkinsonism in the elderly. *Lancet* 1984; 2:1082.
99. Avorn J, Gurwitz JH, Bohn RL, et al. Increased incidence of levodopa therapy following metoclopramide use. *JAMA* 1995; 274:1780.
100. Gill SS, Mamdani M, Naglie G, et al. A prescribing cascade involving cholinesterase inhibitors and anticholinergic drugs. *Arch Intern Med* 2005; 165:808.
101. Savage RD, Visentin JD, Bronskill SE, et al. Evaluation of a Common Prescribing Cascade of Calcium Channel Blockers and Diuretics in Older Adults With Hypertension. *JAMA Intern Med* 2020; 180:643.
102. Holbrook AM, Pereira JA, Labiris R, et al. Systematic overview of warfarin and its drug and food interactions. *Arch Intern Med* 2005; 165:1095.
103. Juurlink DN, Mamdani MM, Kopp A, et al. Drug-induced lithium toxicity in the elderly: a population-based study. *J Am Geriatr Soc* 2004; 52:794.
104. Shorr RI, Ray WA, Daugherty JR, Griffin MR. Concurrent use of nonsteroidal anti-inflammatory drugs and oral anticoagulants places elderly persons at high risk for hemorrhagic peptic ulcer disease. *Arch Intern Med* 1993; 153:1665.
105. Battistella M, Mamdani MM, Juurlink DN, et al. Risk of upper gastrointestinal hemorrhage in warfarin users treated with nonselective NSAIDs or COX-2 inhibitors. *Arch*

- Intern Med 2005; 165:189.
106. Juurlink DN, Mamdani M, Kopp A, et al. Drug-drug interactions among elderly patients hospitalized for drug toxicity. JAMA 2003; 289:1652.
 107. Ray WA, Griffin MR, Schaffner W, et al. Psychotropic drug use and the risk of hip fracture. N Engl J Med 1987; 316:363.
 108. Tamblyn R, Abrahamowicz M, du Berger R, et al. A 5-year prospective assessment of the risk associated with individual benzodiazepines and doses in new elderly users. J Am Geriatr Soc 2005; 53:233.
 109. Garg AX, Papaioannou A, Ferko N, et al. Estimating the prevalence of renal insufficiency in seniors requiring long-term care. Kidney Int 2004; 65:649.
 110. Breton G, Froissart M, Janus N, et al. Inappropriate drug use and mortality in community-dwelling elderly with impaired kidney function--the Three-City population-based study. Nephrol Dial Transplant 2011; 26:2852.
 111. Aronoff GR, Berns JS, Brier ME, et al.. Drug prescribing in renal failure: Dosing guidelines for adults, 4th ed, American College of Clinicians, Philadelphia 1999.
 112. Chertow GM, Lee J, Kuperman GJ, et al. Guided medication dosing for inpatients with renal insufficiency. JAMA 2001; 286:2839.
 113. Gurwitz JH, Field TS, Judge J, et al. The incidence of adverse drug events in two large academic long-term care facilities. Am J Med 2005; 118:251.
 114. Doshi JA, Shaffer T, Briesacher BA. National estimates of medication use in nursing homes: findings from the 1997 medicare current beneficiary survey and the 1996 medical expenditure survey. J Am Geriatr Soc 2005; 53:438.
 115. Gurwitz JH, Field TS, Avorn J, et al. Incidence and preventability of adverse drug events in nursing homes. Am J Med 2000; 109:87.
 116. Gurwitz JH, Field TS, Radford MJ, et al. The safety of warfarin therapy in the nursing home setting. Am J Med 2007; 120:539.
 117. Leipzig RM, Cumming RG, Tinetti ME. Drugs and falls in older people: a systematic review and meta-analysis: I. Psychotropic drugs. J Am Geriatr Soc 1999; 47:30.
 118. Bronskill SE, Anderson GM, Sykora K, et al. Neuroleptic drug therapy in older adults newly admitted to nursing homes: incidence, dose, and specialist contact. J Am Geriatr Soc 2004; 52:749.
 119. Rochon PA, Stukel TA, Bronskill SE, et al. Variation in nursing home antipsychotic prescribing rates. Arch Intern Med 2007; 167:676.
 120. US Food and Drug Administration. FDA MedWatch Safety Alert: RISPERDAL(risperidone), 2003. Available at: www.fda.gov/medwatch/SAFETY/2003/safety03.htm#risper (Accessed on February 09, 2006).

121. www.accessdata.fda.gov/scripts/cdrh/cfdocs/psn/transcript.cfm?show=27#6 (Accessed on February 09, 2006).
122. [Lenzer J. FDA warns about using antipsychotic drugs for dementia. BMJ 2005; 330:922.](#)
123. [Schneider LS, Dagerman KS, Insel P. Risk of death with atypical antipsychotic drug treatment for dementia: meta-analysis of randomized placebo-controlled trials. JAMA 2005; 294:1934.](#)
124. [Wang PS, Schneeweiss S, Avorn J, et al. Risk of death in elderly users of conventional vs. atypical antipsychotic medications. N Engl J Med 2005; 353:2335.](#)
125. [Gill SS, Bronskill SE, Normand SL, et al. Antipsychotic drug use and mortality in older adults with dementia. Ann Intern Med 2007; 146:775.](#)
126. [Onder G, Petrovic M, Tangiisuran B, et al. Development and validation of a score to assess risk of adverse drug reactions among in-hospital patients 65 years or older: the GerontoNet ADR risk score. Arch Intern Med 2010; 170:1142.](#)
127. [Bates DW, Leape LL, Cullen DJ, et al. Effect of computerized physician order entry and a team intervention on prevention of serious medication errors. JAMA 1998; 280:1311.](#)
128. [Peterson JF, Kuperman GJ, Shek C, et al. Guided prescription of psychotropic medications for geriatric inpatients. Arch Intern Med 2005; 165:802.](#)
129. [Gurwitz JH, Field TS, Rochon P, et al. Effect of computerized provider order entry with clinical decision support on adverse drug events in the long-term care setting. J Am Geriatr Soc 2008; 56:2225.](#)
130. [Gurwitz JH, Field TS, Harrold LR, et al. Incidence and preventability of adverse drug events among older persons in the ambulatory setting. JAMA 2003; 289:1107.](#)
131. [Field TS, Mazor KM, Briesacher B, et al. Adverse drug events resulting from patient errors in older adults. J Am Geriatr Soc 2007; 55:271.](#)
132. [Forster AJ, Clark HD, Menard A, et al. Adverse events among medical patients after discharge from hospital. CMAJ 2004; 170:345.](#)
133. [Cochrane RA, Mandal AR, Ledger-Scott M, Walker R. Changes in drug treatment after discharge from hospital in geriatric patients. BMJ 1992; 305:694.](#)
134. [Gurwitz J. Case and commentary. Spotlight case. Double Trouble. Medicine, September 2005. Available at: \[www.webmm.ahrq.gov/case.aspx?caseID=104\]\(http://www.webmm.ahrq.gov/case.aspx?caseID=104\) \(Accessed on March 27, 2006\).](#)
135. [The Institute for Safe Medication Practices. How to Take Your Medications Safely. Available at: <http://www.ismp.org/Newsletters/consumer/alerts/Brochure.asp> \(Accessed on March 27, 2006\).](#)
136. [Cramer JA. Enhancing patient compliance in the elderly. Role of packaging aids and monitoring. Drugs Aging 1998; 12:7.](#)

137. Milton JC, Hill-Smith I, Jackson SH. Prescribing for older people. *BMJ* 2008; 336:606.
138. Bell CM, Fischer HD, Gill SS, et al. Initiation of benzodiazepines in the elderly after hospitalization. *J Gen Intern Med* 2007; 22:1024.
139. Calkins DR, Davis RB, Reiley P, et al. Patient-physician communication at hospital discharge and patients' understanding of the postdischarge treatment plan. *Arch Intern Med* 1997; 157:1026.
140. Boockvar K, Fishman E, Kyriacou CK, et al. Adverse events due to discontinuations in drug use and dose changes in patients transferred between acute and long-term care facilities. *Arch Intern Med* 2004; 164:545.
141. Hajjar ER, Hanlon JT, Sloane RJ, et al. Unnecessary drug use in frail older people at hospital discharge. *J Am Geriatr Soc* 2005; 53:1518.
142. Green JL, Hawley JN, Rask KJ. Is the number of prescribing physicians an independent risk factor for adverse drug events in an elderly outpatient population? *Am J Geriatr Pharmacother* 2007; 5:31.
143. Scott IA, Gray LC, Martin JH, Mitchell CA. Minimizing inappropriate medications in older populations: a 10-step conceptual framework. *Am J Med* 2012; 125:529.
144. Holmes HM, Min LC, Yee M, et al. Rationalizing prescribing for older patients with multimorbidity: considering time to benefit. *Drugs Aging* 2013; 30:655.
145. George CJ, Jacobs LG. Geriatrics medication management rounds: a novel approach to teaching rational prescribing with the use of the medication screening questionnaire. *J Am Geriatr Soc* 2011; 59:138.
146. Munson JC, Bynum JP, Bell JE, et al. Patterns of Prescription Drug Use Before and After Fragility Fracture. *JAMA Intern Med* 2016; 176:1531.
147. Wilson IB, Schoen C, Neuman P, et al. Physician-patient communication about prescription medication nonadherence: a 50-state study of America's seniors. *J Gen Intern Med* 2007; 22:6.
148. Cornish PL, Knowles SR, Marchesano R, et al. Unintended medication discrepancies at the time of hospital admission. *Arch Intern Med* 2005; 165:424.
149. Forman DE, Coletta D, Kenny D, et al. Clinical issues related to discontinuing digoxin therapy in elderly nursing home patients. *Arch Intern Med* 1991; 151:2194.
150. Packer M, Gheorghiade M, Young JB, et al. Withdrawal of digoxin from patients with chronic heart failure treated with angiotensin-converting-enzyme inhibitors. *RADIANCE Study. N Engl J Med* 1993; 329:1.
151. Garfinkel D, Mangin D. Feasibility study of a systematic approach for discontinuation of multiple medications in older adults: addressing polypharmacy. *Arch Intern Med* 2010; 170:1648.

152. Appel LJ, Espeland MA, Easter L, et al. Effects of reduced sodium intake on hypertension control in older individuals: results from the Trial of Nonpharmacologic Interventions in the Elderly (TONE). *Arch Intern Med* 2001; 161:685.
153. Whelton PK, Appel LJ, Espeland MA, et al. Sodium reduction and weight loss in the treatment of hypertension in older persons: a randomized controlled trial of nonpharmacologic interventions in the elderly (TONE). TONE Collaborative Research Group. *JAMA* 1998; 279:839.
154. Griffin MR, Piper JM, Daugherty JR, et al. Nonsteroidal anti-inflammatory drug use and increased risk for peptic ulcer disease in elderly persons. *Ann Intern Med* 1991; 114:257.
155. Gurwitz JH, Soumerai SB, Avorn J. Improving medication prescribing and utilization in the nursing home. *J Am Geriatr Soc* 1990; 38:542.
156. Mamdani M, Juurlink DN, Lee DS, et al. Cyclo-oxygenase-2 inhibitors versus non-selective non-steroidal anti-inflammatory drugs and congestive heart failure outcomes in elderly patients: a population-based cohort study. *Lancet* 2004; 363:1751.
157. Turner MR, Camacho X, Fischer HD, et al. Levothyroxine dose and risk of fractures in older adults: nested case-control study. *BMJ* 2011; 342:d2238.
158. Wolf MS, Curtis LM, Waite K, et al. Helping patients simplify and safely use complex prescription regimens. *Arch Intern Med* 2011; 171:300.
159. Wolf MS, Shekelle P, Choudhry NK, et al. Variability in pharmacy interpretations of physician prescriptions. *Med Care* 2009; 47:370.
160. Thomson MS, Gruneir A, Lee M, et al. Nursing time devoted to medication administration in long-term care: clinical, safety, and resource implications. *J Am Geriatr Soc* 2009; 57:266.
161. Harvath TA, Lindauer A, Sexson K. Managing Complex Medication Regimens. *Am J Nurs* 2016; 116:43.

Topic 3013 Version 79.0

GRAPHICS

Table 1: Selected high-risk drugs

Drug	Potential harm	Comment
Insulin	Hypoglycemia	May often be appropriate; however, aggressive glycemic control may often yield greater harms than benefits in older adults. ^[1-3]
Sulfonylureas	Hypoglycemia	Older hospitalized patients at significant risk for hypoglycemia; avoid or use with great caution. ^[4]
Warfarin	Gastrointestinal, intracranial bleeding	Although a high-risk drug, benefits of warfarin therapy often outweigh harms; maintenance of prothrombin time/international normalized ratio (INR) in therapeutic range tightly linked to risk/benefit ratio. ^[5]
Digoxin	Impairment of cognition, heart block	May have a third-line role in management of systolic heart failure; suboptimal choice for rate control in atrial fibrillation.
Benzodiazepines	Falls	Associated with as much as a 60% increase in fall risk. ^[6]
Diphenhydramine, other first-generation antihistamines	Impaired cognition, urinary retention in men	Poor choice as sleep aid due to anticholinergic effects, next-day sedation, impact on performance including driving; close medication reconciliation important because patients may also obtain over-the-counter drugs.
Opioid analgesics	Constipation, sedation, confusion, cardiorespiratory depression, seizures	Codeine, meperidine, pentazocine, butorphanol, and nalbuphine are poor choices for analgesia. Fentanyl, morphine, or oxycodone are often appropriate with careful dose adjustment.
Antipsychotics	Death, pneumonia	Elevated risk of death when used to treat behavioral complications of dementia, although, in selected cases, benefits may exceed risks if consistent with patient goals of care. ^[7]
Chemotherapeutic agents	Myelosuppression (neutropenia, anemia), hepatotoxicity, cardiotoxicity	Comprehensive assessment is required for determining goals of treatment, particularly in light of comorbidities. When indicated, chemotherapy dose and schedule should be carefully individualized for organ function and anticipated toxicities of treatment. In general, greater treatment-related toxicity is accepted when the expected outcome of treatment is cure.
Selected antimicrobials		
Fluoroquinolones	Tendon inflammation and rupture,	Elevated risk of tendon rupture in combination with glucocorticoids.

	hypoglycemia, cardiac arrhythmias, <i>Clostridioides difficile</i> -associated diarrhea, exacerbation of myasthenia gravis	
Nitrofurantoin	In chronic use (rarely): Pulmonary fibrosis, neuropathy, hepatotoxicity	Avoid in older adults with creatinine clearance <30 mL/minute; does not reach therapeutic concentrations in urine and increased risk of toxicity.
Trimethoprim- sulfamethoxazole (co-trimoxazole)	Hyperkalemia, hypoglycemia (with sulfonylurea), severe dermatologic reaction (rare)	Drug interactions include warfarin (↑ INR), agents that increase serum potassium, and sulfonylureas (↑ hypoglycemic effect).

References:

1. Brown AF, Mangione CM, Saliba D, Sarkisian CA. California Healthcare Foundation/American Geriatrics Society panel on improving care for elders with diabetes. Guidelines for improving the care of the older person with diabetes mellitus. *J Am Geriatr Soc* 2003; 51:S265.
2. Greenfield S, Billimek J, Pellegrini F, et al. Comorbidity affects the relationship between glycemic control and cardiovascular outcomes in diabetes. *Ann Intern Med* 2009; 151:854.
3. Gerstein HC, Miller ME, Byington RP, et al. Action to Control Cardiovascular Risk in Diabetes Study Group. Effects of intensive glucose lowering in type 2 diabetes. *N Engl J Med* 2008; 358:2545.
4. Deussenberry CM, Coley KC, Korytkowski MT, et al. Hypoglycemia in hospitalized patients treated with sulfonylureas. *Pharmacotherapy* 2012; 32:613.
5. Mant J, Hobbs FD, Fletcher K, et al. BAFTA investigators; Midland Research Practices Network (MidReC). Warfarin versus aspirin for stroke prevention in an elderly community population with atrial fibrillation (the Birmingham Atrial Fibrillation Treatment of the Aged Study, BAFTA). *Lancet* 2007; 370:493.
6. Woolcott JC, Richardson KJ, Wiens MO, et al. Meta-analysis of the impact of 9 medication classes on falls in elderly persons. *Arch Intern Med* 2009; 169:1952.
7. Schneider LS, Dagerman KS, Insel P. Risk of death with atypical antipsychotic drug treatment for dementia. *JAMA* 2005; 294:1934.
8. American Geriatrics Society 2015 Beers Criteria Update Expert Panel. American Geriatrics Society 2015 updated Beers Criteria for potentially inappropriate medication use in older adults. *J Am Geriatr Soc* 2015;63:2227.

Revisions and additional information included with data from: Steinman MA, Hanlon JT. Managing medications in clinically complex elders: "There's got to be a happy medium." *JAMA* 2010; 304:1592.

Graphic 51353 Version 14.0

Table 2: Appropriate prescribing in the elderly

1. Is there an indication for the drug?
2. Is the medication effective for the condition?
3. Is the dosage correct?
4. Are the directions correct?
5. Are the directions practical?
6. Are there clinically significant drug-drug interactions?
7. Are there clinically significant drug-disease/condition interactions?
8. Is there unnecessary duplication with other drugs?
9. Is the duration of therapy acceptable?
10. Is this drug the least expensive alternative compared with others of equal usefulness?

Reproduced from: Hanlon JT, Schmader KE, Samsa GP, et al. A method for assessing drug therapy appropriateness. J Clin Epidemiol 1992; 45:1045. Illustration used with the permission of Elsevier Inc. All rights reserved.

Graphic 64079 Version 2.0

Table 3: Anticholinergic activity of medications

Class	Drugs	Relative anticholinergic potency
Antihistamines	H ₁ receptor antagonists, first-generation: brompheniramine, carbinoxamine, chlorpheniramine, clemastine, cyproheptadine, dimenhydrinate, diphenhydramine, doxepin, doxylamine, hydroxyzine, meclizine, triprolidine, others	High
	H ₁ receptor antagonists, second-generation: fexofenadine, cetirizine [*] , loratadine, desloratadine, levocetirizine, others	Low
Antiparkinson	Benztropine, tropatepine, trihexyphenidyl	High
	Amantadine, bromocriptine, entacapone	Low
Analgesic	Opioids: codeine, hydrocodone, fentanyl, meperidine, methadone, morphine, oxycodone, tramadol, others	Low
Antimuscarinic, overactive bladder	Darifenacin, fesoterodine, flavoxate, oxybutynin, solifenacin, tolterodine, trospium	High
Antimuscarinic, spasmolytic	Atropine, belladonna-containing medications, clidinium-chlordiazepoxide, dicyclomine, hyoscyamine, glycopyrrolate, homatropine, methscopolamine, propantheline, scopolamine (hyoscine)	High
Antimuscarinic, inhaled bronchodilator	Ipratropium, tiotropium	High (local effect)
Antimuscarinic, ophthalmic drops (mydriatic/cycloplegic)	Atropine, cyclopentolate, homatropine, scopolamine	High (local effect)
Cardiovascular	Disopyramide	Low
Gastrointestinal	Antiemetics (eg, hydroxyzine, meclizine, promethazine, scopolamine); also refer to first-generation antihistamines above	High
	Domperidone, loperamide, prochlorperazine	Low
	H ₂ receptor antagonists (cimetidine, famotidine [¶])	Low
Muscle relaxant	Orphenadrine, tizanidine	High
	Cyclobenzaprine [*] , baclofen, methocarbamol	Low
Psychotropic	Antipsychotics, first-generation: chlorpromazine, fluphenazine, loxapine, methotrimeprazine	High

	(levomepromazine), thioridazine, trifluoperazine	
	Antipsychotics, first-generation: haloperidol, perphenazine*, others	Low
	Antipsychotics, second-generation: clozapine	High
	Antipsychotics, second-generation: olanzapine*, quetiapine*, iloperidone, risperidone, others	Low
	Benzodiazepines: chlordiazepoxide, clonazepam, temazepam, triazolam	Low
	Selective serotonin reuptake inhibitor (SSRI) antidepressants: citalopram, fluoxetine, fluvoxamine, paroxetine*	Low
	Tricyclic antidepressants: amitriptyline, clomipramine, desipramine, doxepin, imipramine, nortriptyline, others	High
Other neurologic	Carbamazepine, lithium, nefazodone, oxcarbazepine, phenelzine, trazodone	Low

A large number of medicines are reported to have some anticholinergic activity, and considerable variation exists in potency rankings assigned to specific drugs using available anticholinergic risk scales and in expert lists; this list is not exhaustive. Increasing dose and additive effects from simultaneous use of more than one anticholinergic drug can alter the anticholinergic activity rating provided in this table.

* Classified as moderate or high anticholinergic potency in some references or variable effects reported.

¶ Intravenous famotidine use has been associated with central nervous system effects (eg, delirium and confusion) in hospitalized older adults and patients with renal function impairment; this may be due to a central anticholinergic effect.

Courtesy of Paula A Rochon, MD with additional data from:

1. Durán CE, Azemai M, Vander Stichele RH. Systematic review of anticholinergic risk scales in older adults. *Eur J Clin Pharmacol* 2013; 69:1485.
2. Salahudeen MS, Hilmer SN, Nishtala PS. Comparison of anticholinergic risk scales and associations with adverse health outcomes in older people. *J Am Geriatr Soc* 2015; 63:85.
3. United States prescribing information available at US National Library of Medicine DailyMed website (<https://dailymed.nlm.nih.gov/dailymed/>).
4. American Geriatrics Society 2015 Beers Criteria Update Expert Panel. American Geriatrics Society 2015 updated Beers Criteria for potentially inappropriate medication use in older adults. *J Am Geriatr Soc* 2015; 63:2227.

Graphic 100289 Version 16.0

Table 4: Quality indicators for appropriate medication use in vulnerable older adults

Indicator title	Description	Rationale
Medication list	An up-to-date medication list that includes over-the-counter medications should be accessible to all health care providers in the medical record.	Enables identification of potential drug-related causes of new symptoms, eliminates inappropriate duplications, allows review for drug-drug interactions, and allows streamlining of regimen to improve adherence.
Annual drug regimen review	All vulnerable older adults should have an annual drug regimen review.	Allows an opportunity for discontinuing unnecessary medications, or addition of necessary drugs that are not currently prescribed.
Drug indication	All drugs prescribed for vulnerable older adults should have a clearly defined indication.	Allows discontinuing medications that may have been prescribed for unclear or transient indications.
Patient education	All vulnerable older adults (or caregivers) should receive appropriate education about the use of any prescribed drug.	Education may improve adherence and clinical outcomes; also can alert patients or caregivers to potential adverse effects.
Response to therapy	Response to therapy should be documented for all ongoing medical conditions.	Documenting response will help clarify whether a drug is meeting the therapeutic goal for which it was prescribed and provides a basis for continuation, modification, or discontinuation.
Education for warfarin therapy	Patients newly prescribed warfarin should receive education about diet, drug interactions, and risk of bleeding, or should be referred to an anticoagulation clinic.	Awareness of drugs and dietary substances that interact with warfarin can decrease the risk of bleeding complications.
Monitoring warfarin therapy	When warfarin is prescribed, INR should be determined within 4 days of initiation of therapy and at least every 6 weeks thereafter.*	Older adults are at high risk for drug toxicity, and close monitoring can help maintain the INR within the therapeutic range.
Monitoring ACE inhibitor therapy	When ACE inhibitor therapy is prescribed, a serum creatinine and potassium should be monitored within 2 weeks after initiation of therapy and at least yearly thereafter.	Older adults are at increased risk of renal insufficiency and hyperkalemia.
Monitoring loop diuretic therapy	When loop diuretic therapy is prescribed, electrolytes should be	Risk of hypokalemia due to diuretic therapy.

	checked within 2 weeks after initiation and at least annually.	
Avoid propoxyphene	Do not prescribe propoxyphene as an analgesic agent.	Propoxyphene is inferior to, or at best equivalent to, acetaminophen or other analgesics with better safety profiles.
Avoid chronic or high-dose benzodiazepine use	If a benzodiazepine is taken for more than 1 month, there should be documentation of discussion of risks and attempt to taper or discontinue.	Benzodiazepines increase the risk of falls, hip fracture, and confusion.
Avoid drugs with strong anticholinergic properties	Do not prescribe drug therapies with a strong anticholinergic effect if alternative therapies are available.	These therapies are associated with adverse events such as confusion, urinary retention, constipation, visual disturbance, and hypotension.
Avoid barbiturates	If an older adult does require the therapy for control of seizures, do not use barbiturates.	These therapies are potent central nervous system depressants, have a low therapeutic index, are highly addictive, cause drug interactions, and are associated with an increased risk for falls and hip fracture.
Avoid meperidine as an opioid analgesic	When analgesia is required, avoid use of meperidine.	This therapy is associated with an increased risk for delirium and may be associated with the development of seizures.
Avoid chronic use of ketorolac	Ketorolac should not be prescribed for more than 5 days.	This therapy is associated with a high risk of GI side effects, including bleeding, and other analgesics are safer in older patients.
Avoid skeletal muscle relaxants	Skeletal muscle relaxants (cyclobenzaprine, methocarbamol, carisoprodol, chlorzoxazone, orphenadrine, tizanidine, metaxalone) should not be prescribed for more than 1 week.	These medications can cause anticholinergic adverse effects, sedation, confusion, and data of efficacy are limited.
Avoid ticlopidine	Clopidogrel should be prescribed rather than ticlopidine for patients who require antiplatelet therapy (eg, recent stroke, myocardial infarction, acute coronary syndrome, percutaneous angioplasty).	Ticlopidine may be less effective than clopidogrel and is associated with a higher risk of hematological disorders than clopidogrel.
Treat iron deficiency anemia with low-dose oral iron therapy	Vulnerable older adults with iron deficiency anemia should take no more than 1 low-dose oral iron tablet daily.	Low-dose therapy is equally effective with fewer adverse effects than high-dose oral iron therapy.
Antipsychotic medication	An assessment of response should be documented within 1 month for older	The use of antipsychotic drugs increases mortality in older adults, and

response	adults started on an antipsychotic drug.	behavioral modification is an effective alternative.
Acetaminophen	Older adults prescribed high-dose (≥ 3 g per day) acetaminophen, or those with liver disease taking acetaminophen chronically, should be advised of the risk of liver toxicity.	The risk of liver toxicity is greater with use of acetaminophen.
NSAIDs and aspirin	The risk of GI bleeding should be discussed and documented. Individuals at increased risk for GI bleeding (age >75 years, peptic ulcer disease, history of GI bleeding, warfarin use, chronic glucocorticoid use) should be treated concomitantly with misoprostol or a proton pump inhibitor when treated with a nonselective NSAID.	Risks of GI bleeding are increased in older adults taking NSAIDs or daily aspirin.

INR: international normalized ratio; ACE: angiotensin-converting enzyme; GI: gastrointestinal; NSAIDs: nonsteroidal antiinflammatory drugs.

* These monitoring frequencies represent minimal indications; many experts would advise daily monitoring initially and monitoring every four weeks once a stable and at-goal therapeutic INR has been achieved.

Information from: Shrank WH, Polinski JM, Avorn J. Quality indicators for medication use in vulnerable elders. J Amer Geriatr Soc 2007; 55:S373.

Graphic 73834 Version 9.0

Table 5: Examples of prescribing cascades

Initial drug therapy	Adverse drug event	Subsequent drug therapy
Antipsychotics	Extrapyramidal signs and symptoms	Antiparkinsonian therapy
Cholinesterase inhibitors	Urinary incontinence	Incontinence treatment
Thiazide diuretics	Hyperuricemia	Gout treatment
NSAIDs	Increased blood pressure	Antihypertensive therapy

Medication prescribing cascades occur when patients are prescribed medications to treat the adverse side effects of previously prescribed medications. This leads to polypharmacy and further increases the risk for adverse drug events. Periodic review of medication lists, especially in older adults, can minimize this risk.

NSAIDs: nonsteroidal antiinflammatory drugs.

Data from:

1. Rochon PA, Gurwitz JH. Optimizing drug therapy for elderly people: the prescribing cascade. *BMJ* 1997; 315:1096.
2. Gill SS, Mamdani M, Naglie G, et al. A prescribing cascade involving cholinesterase inhibitors and anticholinergic drugs. *Arch Intern Med* 2005; 165:808.

Graphic 74478 Version 5.0

Table 6: Frequency of adverse drug events and preventable adverse drug events by drug class

Drug class	Total adverse drug events (n = 815) N (percent)	Preventable adverse drug events (n = 338) N (percent)
Warfarin	121 (15)	42 (12)
Atypical antipsychotics	92 (11)	42 (12)
Loop diuretics	69 (8)	33 (10)
Opioids	51 (6)	26 (8)
Antiplatelets	46 (6)	23 (7)
ACE inhibitors	45 (6)	27 (8)
Antidepressants	43 (5)	25 (7)
Laxatives	43 (5)	16 (5)
Benzodiazepines (intermediate acting)	39 (5)	30 (9)
Insulins	37 (5)	18 (5)

Only drug classes with the frequency of adverse drug events of 5 percent and more are presented. Some adverse drug events were associated with more than one drug class.

ACE: angiotensin-converting enzyme.

Adapted with permission from: Gurwitz, MD, Field, T, Judge, J, Rochon, P, et al. The incidence of adverse drug events in two large academic long-term care facilities. Am J Med 2005; 118:251. Copyright © 2005 Excerpta Media.

Graphic 71281 Version 2.0

Table 7: Frequency of adverse drug events by type

Type	Total adverse drug events (n = 815) N (percent)	Preventable adverse drug events (n = 338) N (percent)
Neuropsychiatric	199 (24)	97 (29)
Hemorrhagic	159 (20)	53 (16)
Gastrointestinal	140 (17)	55 (16)
Renal/electrolytes	80 (10)	40 (12)
Metabolic/endocrine	64 (8)	35 (10)
Cardiovascular	36 (4)	15 (4)
Dermatologic	36 (4)	4 (1)
Extrapyramidal symptoms	30 (4)	7 (2)
Fall with injury	21 (3)	17 (5)
Fall without injury	21 (3)	11 (3)
Infection	19 (2)	1 (<1)
Syncope/dizziness	16 (2)	8 (2)
Anticholinergic	9 (1)	3 (1)
Ataxia/difficulty with gait	9 (1)	5 (2)
Hematologic	8 (1)	3 (1)
Respiratory	6 (1)	4 (1)
Anorexia	3 (<1)	2 (<1)
Functional decline	3 (<1)	2 (<1)
Hepatic	1 (<1)	1 (<1)

Adverse drug events could manifest as more than one type. Neuropsychiatric events include oversedation, confusion, hallucinations, and delirium. Anticholinergic effects include dry mouth, dry eyes, urinary retention, and constipation.

Reproduced with permission from: Gurwitz, MD, Field, T, Judge, J, Rochon, P, et al. The incidence of adverse drug events in two large academic long-term care facilities. Am J Med 2005; 118:251. Copyright © 2005 Excerpta Media.

Graphic 72518 Version 3.0

Table 8: A step-wise approach to reviewing medications for older adults

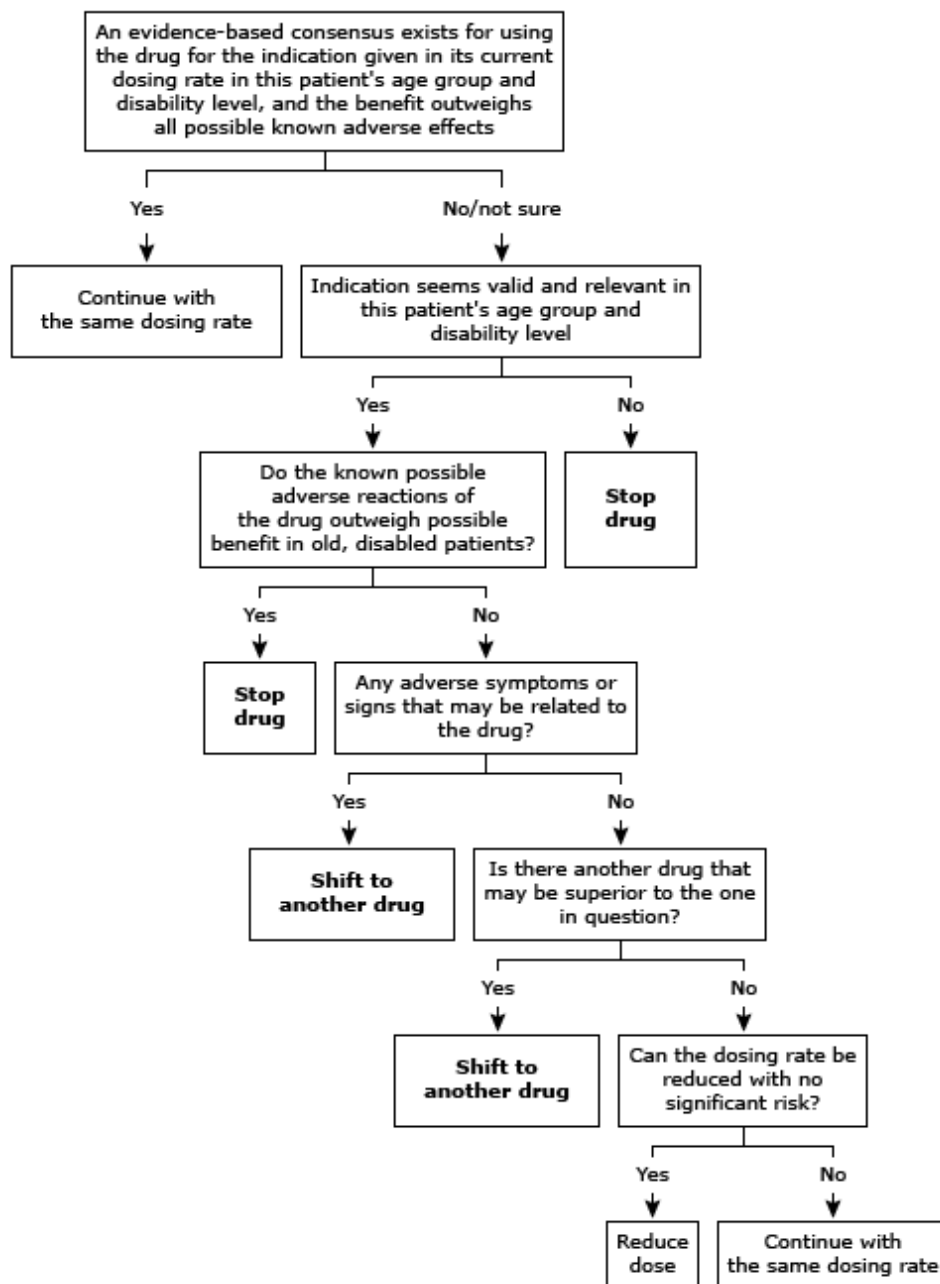
Approach
Review current drug therapy
Discontinue potentially unnecessary therapy
Consider adverse drug events as a potential cause for any new symptom
Consider nonpharmacologic approaches
Substitute with safer alternatives
Reduce the dose
Use beneficial therapies when indicated

Reproduced with permission from: Rochon PA, Gurwitz JH. Drug Therapy. The Lancet 1995; 346:32. Copyright © 1995 Elsevier.

Graphic 52450 Version 4.0

Algorithm 1: Medication review

Discuss the following with the patient/guardian



Reproduced with permission from: Garfinkel D, Mangin D. Feasibility study of a systematic approach for discontinuation of multiple medications in older adults: Addressing polypharmacy. *Arch Intern Med* 2010; 170:1648. Copyright © 2010 American Medical Association. All rights reserved.

Graphic 61655 Version 13.0

Contributor Disclosures

Paula A Rochon, MD, MPH, FRCPC No relevant financial relationship(s) with ineligible companies to disclose. **Kenneth E Schmader, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Jane Givens, MD, MSCE** No relevant financial relationship(s) with ineligible companies to disclose.

Contributor disclosures are reviewed for conflicts of interest by the editorial group. When found, these are addressed by vetting through a multi-level review process, and through requirements for references to be provided to support the content. Appropriately referenced content is required of all authors and must conform to UpToDate standards of evidence.

[Conflict of interest policy](#)

→